Universitat de Barcelona

Master in Quantum Science and Technology

Master's thesis

Universal properties from quantum many-body dynamics

Joaquín G. Márquez Olguín

Supervisor: Stefano Carignano

Barcelona Supercomputing Center
Centro Nacional de Supercomputación

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Supervised by: Stefano Carignano

Barcelona Supercomputing Center, Plaça Eusebi Güell, 08034 Barcelona

Aug –, 2026

The Loschmidt echo—the amplitude for a quantum state to return to itself—provides a dynamical route to obtain universal properties of a critical many-body system. Analytically continuing from real to imaginary time, the echo becomes the partition function of a conformal field theory on a strip. The central charge and operator content of the corresponding conformal-field theory are then encoded in the spectrum of a transfer matrix, obtained by contracting the space–time tensor network along the space direction. The cost of this transverse contraction is governed by the temporal entanglement, which grows only logarithmically at criticality, allowing the method to reach times where conventional evolution algorithms are limited by the entanglement barrier. These results were established for integrable chains, and this thesis extends them to the non-integrable regime. We study a transverse-field Ising model extended by next-nearest-neighbour interactions, which remains critical and in the Ising universality class at weak frustration. We encode its evolution as a matrix product operator, and we resolve the leading spectrum of the resulting non-Hermitian transfer matrix with a block power method that evolves its left and right eigenvectors together. Where the method allows, we recover the conformal signatures: the central charge remains consistent with the Ising value $c = \frac{1}{2}$, and the boundary operator content of the Ising class is reproduced. Throughout, we separate the universal quantities from the non-universal lattice constants that accompany them, and we map where and why the method reaches its limits. *This is a working draft.*

Keywords: Tensor Networks, Transverse Contraction, Many-body Systems, Quantum Dynamics, CFT

Joaquín G. Márquez Olguín: jmarquol30@alumnes.ub.edu

Acknowledgements

I thank my dad, my brother, but specially to my mom, my gf Helena, my flatmates...

Note on the use of AI

AI assistants were used during the preparation of this thesis. Anthropic's Claude models assisted with brainstorming, preliminary drafts, debugging numerical code, and automating repetitive tasks, including the production of figures and diagrams. OpenAI's ChatGPT and Google's Gemini assisted with the organisation and clarity of the text and with identifying typographical errors. All AI-assisted material was reviewed, verified, and revised by the author, who takes full responsibility for the content of the thesis.

Contents

Acronyms

CFT Conformal Field Theory . . . 2

DMRG Density Matrix Renormalization Group . . . 1

ETH Eigenstate Thermalization Hypothesis . . . 1

FSM finite-state machine . . . III

JW Jordan–Wigner . . . 10

MPO Matrix Product Operator . . . 6

MPS Matrix Product State . . . 1

NN nearest-neighbour . . . 12

NNN next-nearest-neighbour . . . 2

RDM Reduced Density Matrix . . . 8

RTM Reduced Transition Matrix . . . III

SVD Singular Value Decomposition . . . 29

TDVP Time-Dependent Variational Principle . . . 1

TEBD Time-Evolving Block Decimation . . . 1

TN Tensor Network . . . 1

tMPO temporal Matrix Product Operator . . . 6

tMPS temporal Matrix Product State . . . 15

1 Introduction and Motivation

Following a closed quantum system after being driven far from equilibrium is one of the central questions of modern many-body physics. Simulating such dynamics is difficult, but it can also be revealing: at criticality, the real-time evolution of a simple initial state can expose universal information normally associated with low-energy physics. Our aim with this thesis is to determine how far this dynamical route can be pushed once the system is no longer exactly solvable.

This question is particularly relevant for non-integrable systems, which lack exact analytical solutions. Their dynamics is also intrinsically harder to simulate over long times: after a quench these systems are generically expected to thermalize, acting as their own heat bath so that local observables relax to values fixed only by the energy—the content of the Eigenstate Thermalization Hypothesis (ETH) [1–3]—whereas integrable models, protected by extensively many conserved charges, retain memory of their initial state instead [4]. Analytical results are largely confined to those integrable limits, so almost everything we know about thermalization, scrambling of the quantum information and the onset of chaos in non-integrable models depends on our ability to simulate their dynamics numerically.

Exact numerical simulations are severely limited by the exponential scaling of the Hilbert space, which makes representing a generic many-body wavefunction intractable. To circumvent this bottleneck, the community has turned to Tensor Networks (TNs), a framework that efficiently compresses quantum states by using entanglement as the organizing principle. For one-dimensional systems in equilibrium, Matrix Product States (MPSs) combined with algorithms such as the Density Matrix Renormalization Group (DMRG) have become the state-of-the-art approach [5–7]. Their efficiency comes from the area law of entanglement, which holds for ground states of one-dimensional systems with local gapped Hamiltonians: the entanglement entropy across a bipartition saturates to a constant rather than growing with the subsystem size [8]. Since the required MPS bond dimension is controlled by this entanglement, the ansatz efficiently targets the low-entanglement corner of Hilbert space where these ground states lie.

Simulating dynamics, however, runs into the so-called entanglement barrier. After a quantum quench, the entanglement entropy typically grows linearly in time, approaching volume-law behaviour [9]. Although highly optimised algorithms such as Time-Evolving Block Decimation (TEBD) [10] and Time-Dependent Variational Principle (TDVP) [11] are effective at short times, capturing this growth requires a bond dimension that increases exponentially with time, making conventional simulations intractable at late times [12, 13].

To overcome this barrier, transverse contraction is a framework that adopts a different perspective [14, 15]. Rather than evolving a state forward in the Schrödinger picture, it represents the full dynamics as a two-dimensional tensor network and contracts it along the original spatial direction. The computational cost is then controlled by the temporal entanglement entropy, defined through a Schmidt decomposition across a cut in time rather than in space. Although its physical interpretation remains an open question, it is computationally well defined. For a generic quench it may grow as strongly as spatial entanglement; nevertheless, the transverse viewpoint can provide a useful way to tackle specific dynamical quantities.

The central quantity we compute throughout is the Loschmidt echo for an infinite, translation-invariant system: the amplitude for a state to return to itself after evolving for a time T [16]. It measures how quickly the evolved state departs from, and how coherently it returns towards, its initial condition, providing a natural measure of relaxation after a quench. Given its structure, transverse contraction offers an efficient way to compute it by contracting the space–time network directly with a power method [17].

At a conformally invariant critical point, the echo also provides direct access to universal data such as the central charge of the underlying Conformal Field Theory (CFT). In the transverse picture, it is governed by a spatial transfer matrix whose spectrum is predicted by the CFT and which becomes effectively unitary at long times—a phenomenon known as emergent dual unitarity [18]. This behaviour has been demonstrated explicitly for the critical Ising and three-state Potts chains [17], together with the finite-time corrections [19] that we use in Section 7.

The central question is therefore whether these conformal signatures survive the loss of integrability of the model. To address it, we study a self-dual quantum spin chain in the thermodynamic limit that extends the transverse-field Ising model with next-nearest-neighbour (NNN) interactions controlled by a coupling p [20]. For any $p > 0$ it is non-integrable and longer-ranged, making its dynamics costly for conventional methods, yet it remains critical and in the Ising universality class over a wide range of p [20], so the CFT predictions provide a clear target for numerical simulations. We prepare the system in the fully polarized paramagnet state, quench it to the critical point, and use the transverse contraction machinery to compute the Loschmidt echo and the generalized temporal entropies, which we compare with the CFT predictions.

The rest of this thesis is organized as follows. Section 2 defines the Loschmidt echo and develops the CFT framework that relates the transfer-matrix spectrum and generalized temporal entropy to the central charge and operator content. Section 3 develops the transverse-contraction framework: the rotated picture, the temporal states, and the transition matrix from which the generalized temporal entropies are built. Section 4 introduces the ANNNI-type model, establishes its critical Ising universality, and measures the frustration-dependent sound velocity. Section 5 constructs the time-evolution MPO and its non-Hermitian transfer matrix. Section 6 presents the block power method, and Section 7 validates the numerical results with the CFT predictions.

2 Universal Data from the Dynamics

The real-time dynamics of a critical lattice model contains universal information that can be extracted from a simulation. At a continuous critical point, the lattice model forgets its microscopic details and is described by a CFT—a continuum theory so constrained by symmetry that its predictions are fixed by a small set of numbers: the central charge c and the spectrum of scaling dimensions of its operators. These quantities are universal: every model in the same universality class reproduces them, which is exactly what makes them good targets for a numerical method. This section relates the Loschmidt echo to a boundary CFT, states the resulting universal predictions, and identifies the non-universal lattice parameters needed to test them.

2.1 From the Loschmidt echo to a boundary CFT

The object we compute is the Loschmidt echo,

$$\mathcal{L}(T) = \langle \Psi_0 | \, e^{-iHT} \, | \Psi_0 \rangle \, . \tag{1}$$

For an infinite, translation-invariant chain prepared in $|\Psi_0\rangle$, this is the amplitude to return to the initial state after evolving for a time T with the critical Hamiltonian. We take $|\Psi_0\rangle$

to be a product state; its particular choice determines the boundary condition discussed below.

To connect this amplitude with the continuum theory, we rotate the evolution time to the imaginary axis, $T \to -i\tau$ (a Wick rotation [21]). The real-time propagator e^{-iHT} then becomes the Euclidean weight $e^{-\tau H}$, the same operator that generates the thermal partition function of a two-dimensional statistical system. The matrix element in Eq. (1) can consequently be written as a Euclidean path integral on a slab of width τ, with both edges fixed by $|\Psi_0\rangle$. A state that can terminate a path integral in this way, without introducing any scale of its own, is what a CFT calls a conformal boundary state. The echo is therefore the partition function of the critical theory on a long strip whose two edges carry the boundary condition set by $|\Psi_0\rangle$ and whose width is the (imaginary) evolution time. The rotation is used only to obtain the conformal predictions, which are derived on the Euclidean strip; they are continued back to real time before being compared with the simulation. Figure 1 shows this construction, first as a tensor network and then as the strip it represents.

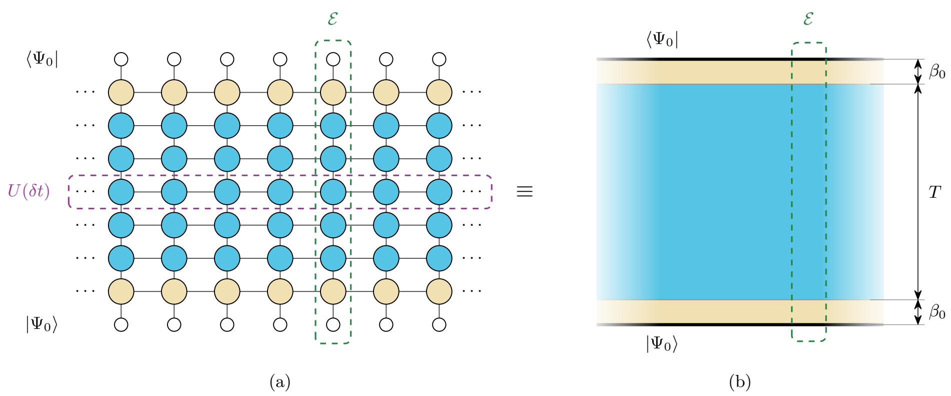


Figure 1: **The Loschmidt echo as a tensor network and as a conformal strip.** (a) Trotterising the evolution expresses the echo as the contraction of a two-dimensional network. Open circles are the product state $|\Psi_0\rangle$ closing the network at either end; each row of filled circles is one Trotter layer $U(\delta t)$, and the cream rows at top and bottom represent the imaginary-time cooling β_0 that brings the bare product state towards a conformal boundary condition. (b) The same object after the Wick rotation, in the continuum. The heavy lines are the edges where $|\Psi_0\rangle$ closes the path integral, the cream bands are the cooling intervals and the blue bulk is the real-time evolution, so the strip spans $T + 2\beta_0$ in the time direction and is infinite in the spatial one.

Two features of this construction are important for the numerics. First, a bare product state carries weight on high-energy configurations and is not itself a conformal boundary state. We therefore cool it in imaginary time for a short interval β_0, which suppresses this high-energy content and brings the state into the regime where the CFT applies. On the lattice, the cooling contributes $N_\beta = 2\beta_0/\delta t$ imaginary-time sites split between the two ends, so the strip spans $T + 2\beta_0$ in the time direction. Second, the initial state is the boundary condition of the strip. Each $|\Psi_0\rangle$ realizes a particular conformal boundary condition, and that choice selects which tower of boundary operators appears in the spectrum [19]. In our convention, we quench from the free boundary, so the Ising class admits the dimensions $\{0, \frac{1}{2}, \frac{3}{2}, 2, \dots\}$; a different choice of initial and final states selects a different tower. The lightest dimension also controls how quickly finite-size corrections decay, so the free boundary at $x_1 = \frac{1}{2}$ converges quickly and the fixed boundary at $x_1 = 2$ slowly. Appendix G measures both conditions at the integrable point.

We contract the strip in the transverse direction. In the Euclidean description, space and imaginary time are coordinates of the same two-dimensional geometry, so either can be chosen as the direction of propagation. In the transverse picture, we exchange their roles and contract the network along the original spatial direction. This defines a transfer matrix $\mathcal{E}$ that propagates from one spatial column to the next while acting on states supported on a temporal slice. Its tensor-network construction is developed in Section 3.

In two dimensions, conformal symmetry strongly constrains both the spectrum of $\mathcal{E}$ and the entanglement across a cut in the strip [22, 23]. These two observables provide independent routes to the universal data; Appendix A.1 summarises the conformal mappings used to derive them.

2.2 The universal predictions

Confining a CFT to an infinite strip of transverse size β shifts its ground-state energy by an amount set by the central charge. The excited levels above this shift are organised by the operator scaling dimensions x_i. The transfer matrix inherits this structure [17],

$$\mathcal{E} = e^{-a\,\hat{H}} = \exp\Big[-\big(-\kappa + \pi L_0\big)\,g\Big], \qquad \kappa = \frac{\pi c}{24}, \tag{2}$$

where L_0 is the operator whose eigenvalues list the scaling dimensions, κ is the Casimir number, and $g = a/\beta$ is the ratio of the column spacing to the strip width, with $a = 1$ in lattice units. The width is complex, $\beta = v(2\beta_0 + iT)$, because the regulator contributes imaginary time while the quench contributes real time. Substituting this width and expanding for $\beta_0 \ll T$ shows that the universal contributions enter the eigenvalue phases at order $1/T$ but affect their moduli only at order β_0/T^2 (Appendix A.2). A single diagonalisation can therefore determine both c and the dimensions x_i from the phases.

The second route uses entanglement. In a CFT, the Rényi entropies follow from correlators of twist operators whose scaling dimension is proportional to c [24]. In the transverse setting, the cut runs across the strip in the time direction, and its two sides do not belong to a single quantum state. The entropies are instead built from the left and right temporal MPSs of Section 3; they are therefore generalized entropies [25] and are generally complex. For a cut at time t [17, 19],

$$S_n^{\text{gen}}(t) = s_0 + \frac{i\pi c}{24}\Big(1+\frac{1}{n}\Big) + \frac{c}{12}\Big(1+\frac{1}{n}\Big)\,W(t,T), \tag{3}$$

with the chord variable

$$W(x,\ell) \;=\; \log\!\left[\frac{2\ell}{\pi}\,\sin\frac{\pi x}{\ell}\right], \tag{4}$$

where x is the position of the cut and ℓ is the extent of the system. Evaluating W on the interval $(0, T)$ makes the evolution time play the role of the subsystem size in the spatial formula. The entropy therefore grows as $\log T$, and the bond dimension required for the contraction grows only polynomially with T [12, 13]. We use the Rényi-2 member throughout because it is the least expensive to compute and requires only two copies of the system, making it experimentally accessible [26]. The equilibrium calculations of Section 4 instead use the ground-state von Neumann entropy, which DMRG provides directly from the entanglement spectrum.

Equations (2) and (3) give four predictions, all tested in Section 7. Three concern the transfer eigenvalues μ_i. We work with their logarithms, $\lambda_i = \log(-\mu_i)$, so that $\operatorname{Im}\lambda_i = \arg(-\mu_i)$ is the phase of μ_i measured from the negative real axis. The minus sign shifts every phase by the same constant π; this shift cancels in phase differences and is absorbed by a constant when fitting $\operatorname{Im}\lambda_0$ alone.

The generalized-entropy profile. Separating Eq. (3) into real and imaginary parts gives

$$\operatorname{Re} S_n^{\text{gen}}(t) = s_0 + \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\, W(t,T), \qquad \operatorname{Im} S_n^{\text{gen}} = \frac{\pi c}{24}\Big(1 + \frac{1}{n}\Big). \tag{5}$$

The real part follows a symmetric chord profile. Plotted against $W(t,T)$, it becomes a straight line whose slope is $\frac{c}{12}(1+1/n)$, derived in Appendix A.1, and therefore determines c. The imaginary part is independent of t and gives a second estimate of c without fitting the spatial profile. Finite-time corrections to S_n^{gen} decay as $T^{-1/n}$ [19]; for the Rényi-2 entropy used here this is $T^{-1/2}$, so extracting c requires extrapolation to the long-time limit.

The central charge from the leading eigenvalue. The central-charge and boundary-gap relations are expansions in $1/T$ of the logarithms λ_i, rather than of the eigenvalues themselves. Substituting the strip width into Eq. (2) and expanding gives the following relation [17], derived in Appendix A.2:

$$\frac{\operatorname{Im} \lambda_0}{T} = a_0 + \frac{B}{T} + \frac{C}{T^2} + \frac{a_4}{T^4} + \cdots\,, \qquad C = -\frac{\kappa}{v} = -\frac{\pi c}{24\, v}. \tag{6}$$

Here a_0 and a_4 are non-universal coefficients and $B = -\pi$ is the branch constant introduced by $\lambda_0 = \log(-\mu_0)$, while the coefficient of $1/T^2$ carries the Casimir number divided by the velocity. Once v has been measured independently (Section 4.2), the relation $c = 24\, v\, |C|/\pi$ determines the central charge from the eigenvalue phase alone, without relying on the entropy. Because this term is small, the extraction requires a nonzero β_0 and sufficiently long times. In practice we fit $\operatorname{Im} \lambda_0 = A\,T + B + C/T + D/T^3$ on the unwrapped phase, where B absorbs the branch shift.

The boundary exponent from the first gap. The gap between the leading and first subleading eigenvalues measures the lightest boundary dimension,

$$\operatorname{Im}(\lambda_1 - \lambda_0) = \frac{\pi x_1}{v\,T}. \tag{7}$$

Once the sound velocity is known, this phase gap determines x_1. More generally, each subleading eigenvalue corresponds to a boundary dimension x_i through the same relation. We refer to this spectrum—the dominant eigenvalue followed by one subleading eigenvalue for each boundary operator, ordered by increasing phase distance—as the conformal tower, and to its members as tower states.

Emergent dual unitarity. Equation (2) also fixes how the spectrum evolves with T. Unlike the two predictions above, this one is an expansion of the eigenvalues themselves rather than of their logarithms: the moduli of the leading eigenvalues approach a common value while their phases stay distinct,

$$\frac{|\mu_i(T)|}{|\mu_0(T)|} = 1 - \frac{a_i}{T} - \frac{b_i}{T^2} - \cdots \xrightarrow[T\to\infty]{} 1. \tag{8}$$

Equal moduli with distinct phases indicate that the temporal transfer matrix approaches a unitary operator up to an overall non-universal normalisation. We test this behaviour by following the dominant eigenvalue $\mu_0(T)$ in the complex plane as it moves along a trajectory of nearly constant radius with a winding phase.

These predictions apply to the continuum theory, whereas a simulation contains additional lattice scales and numerical parameters. The sound velocity v, the entropy offset s_0 and the overall normalisation of the transfer matrix are fixed by the lattice construction rather than by the universality class. We measure v independently in Section 4.2, absorb s_0 into the entropy fits, and remove the normalisation by working with phase differences, so that only universal quantities are compared in Section 7. Section 3 constructs the transfer and transition matrices used to measure them.

3 Time Evolution via Transverse Contraction

Section 2 showed that the Loschmidt echo of a critical quench encodes conformal data in the spectrum of a transfer matrix acting along the spatial direction of the corresponding strip. We now construct this transfer matrix and explain how it is used numerically. Transverse contraction was introduced for infinite chains by representing the dynamics with MPS extending along the time direction [14, 15], and was subsequently developed to study temporal entropies [17, 27]. Our calculations use the `ITransverse.jl` implementation developed in [28].

We rotate the space–time tensor network and contract it along the spatial direction. This reformulation maps the contraction to an eigenproblem for the spatial transfer matrix, whose dominant eigenvectors define the temporal boundary states. These states determine the RTM used to compute the generalized temporal entropies. We finally compare two truncation schemes for controlling their bond dimension.

3.1 The rotated network

In the transverse contraction framework, the time evolution of a D-dimensional system is represented by a $(D+1)$-dimensional tensor network, with the additional dimension corresponding to time. For the one-dimensional chain studied here, the result is a two-dimensional network with no open physical legs. The echo is a scalar amplitude, so this network is analogous to the partition function of a two-dimensional classical model. Its exact contraction is #P-hard in general [29], which requires us to use an approximate representation with a controlled bond dimension (Section 3.4).

The tensor-network representation of the echo follows directly from Eq. (1) and is pictured in Figure 1a. The initial and final states are represented as MPSs, while the evolution operator is decomposed into a sequence of Matrix Product Operator (MPO) layers, one for each Trotter step. Because we require the Hamiltonian to be translation invariant and time independent, the same bulk tensor W appears at every spatial site and time step. Section 5 constructs this tensor for the model with NNN interactions.

In conventional evolution algorithms, successive MPO layers of the time-evolution operator $U(\delta t)$ are applied to the initial MPS $|\Psi_0\rangle$. The resulting spatial entanglement grows linearly with time (volume law) and eventually makes the state intractable at fixed bond dimension [9, 12, 13]. Transverse contraction instead contracts the same network column by column along the spatial direction. The tensors remain unchanged, but their indices exchange roles: the virtual spatial bonds become physical temporal indices, while the physical spatial indices become virtual temporal bonds. Figure 2 illustrates this exchange. The bond dimension of $U(\delta t)$ therefore sets the physical dimension of the temporal problem. Each bulk column defines a temporal Matrix Product Operator (tMPO) $\mathcal{E}$ that acts as a transfer matrix along the original spatial direction, and translation invariance ensures that the same $\mathcal{E}$ is applied at every site.

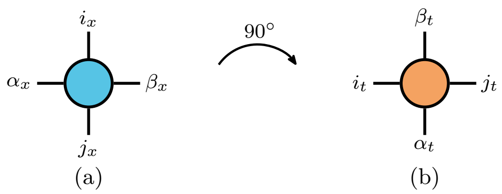


Figure 2: **The rotation exchanges physical and virtual indices.** A rank-4 tensor read (a) along space, with (i_x, j_x) physical and (α_x, β_x) virtual, and (b) along time, where the roles are exchanged.

These objects provide the lattice representation of the strip geometry introduced in Section 2. The strip is infinite along the spatial direction, and applying infinitely many columns on either side of a cut projects the states $\langle L|$ and $|R\rangle$ onto the dominant left and right eigenvectors of $\mathcal{E}$. The resulting fixed points are independent of how the chain is terminated far from the cut. They encode the temporal boundary condition imposed by $|\Psi_0\rangle$ at the top and bottom edges. The strip width is set by the number of temporal sites, including the real-time evolution and the cooling sites associated with the regulator β_0. The CFT constrains the spectrum of the column transfer matrix $\mathcal{E}$, while the overlap $\langle L|R\rangle$ gives the Loschmidt echo after the spatial contraction.

The row and column transfer operators have different spectral properties. Each row represents the unitary Trotter step $U(\delta t) = e^{-iH\delta t}$ and therefore has eigenvalues of unit modulus. By contrast, the column transfer matrix $\mathcal{E}$ is non-Hermitian and generally has complex eigenvalues with different moduli. The implications of its non-Hermiticity for the left and right eigenvectors are discussed in Section 5 after the operator has been constructed explicitly.

3.2 The dynamics as an eigenvalue problem

After the rotation and Trotter decomposition, the two-dimensional network in Figure 1a is contracted along the spatial direction by repeatedly applying the column transfer matrix between the temporal boundaries. For a chain of N spatial sites,

$$\mathcal{L}(T) \;=\; \langle L|\; \mathcal{E}(T)^N \;|R\rangle \;=\; \sum_i c_i\, \mu_i(T)^N \;=\; c_0\, \mu_0(T)^N \Big[\, 1 + \sum_{i\geq 1} \frac{c_i}{c_0} \Big(\frac{\mu_i}{\mu_0}\Big)^N \,\Big], \tag{9}$$

where μ_i are the eigenvalues of $\mathcal{E}$ and the coefficients c_i contain the corresponding overlaps with the boundary states. If μ_0 has the largest modulus, every other contribution is suppressed relative to it by $(|\mu_i|/|\mu_0|)^N$. The thermodynamic limit is therefore determined by the leading eigenvalue, and the intensive Loschmidt rate becomes

$$\ell(T) = -\lim_{N\to\infty} \frac{1}{N} \log |\mathcal{L}(T)| = -\log |\mu_0(T)|, \tag{10}$$

up to $\mathcal{O}(1/N)$ boundary corrections for a finite open chain. The calculation thus reduces to a spectral problem whose cost is independent of the spatial length.

Equation (9) also identifies the spectral data required for the conformal analysis. The modulus of μ_0 determines the Loschmidt rate, its phase encodes the central charge, and the first spectral gap gives access to the boundary operator content. The generalized temporal entropies instead depend on the dominant left and right eigenvectors. Our calculations must therefore resolve the leading eigenpairs of the non-Hermitian operator $\mathcal{E}$.

To obtain these eigenpairs, we implement the block power method described in Section 6. It evolves a block of k temporal states and resolves the k leading eigenvalues together with their corresponding left and right eigenvectors.

Each iteration increases the bond dimension of the temporal states and must therefore be followed by a truncation. The available truncation schemes are introduced in Section 3.4.

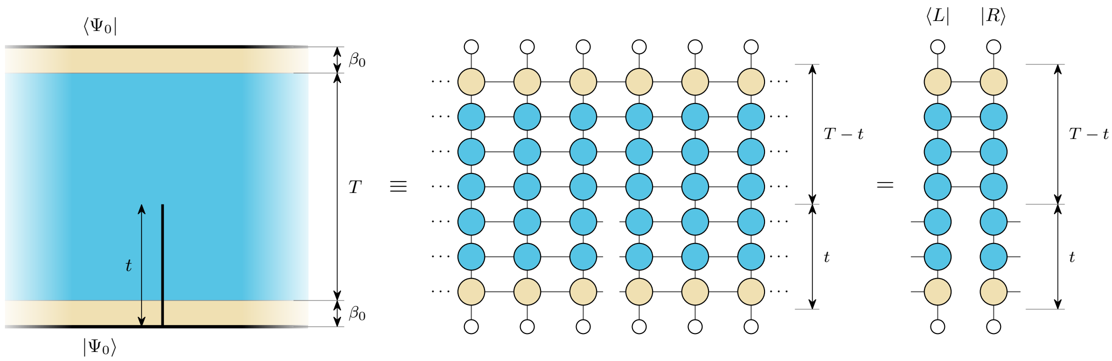
3.3 Temporal states and the RTM

Each application of $\mathcal{E}$ increases the bond dimension of the temporal boundary states. The computational cost is therefore controlled by their temporal entanglement, defined across a cut in the time direction. Section 2.2 showed that in CFT the generalized temporal entropies grow logarithmically with the evolution time. We now define the object from which these entropies are computed.

The dominant left and right fixed points of the non-Hermitian transfer matrix are distinct, and the echo depends on their overlap $\langle L|R\rangle$. An Reduced Density Matrix (RDM) formed from either state alone therefore does not contain the required information. The relevant object is instead the RTM constructed jointly from the two temporal boundaries,

$$\mathcal{T}_t = \frac{\mathrm{Tr}_{>t}\, |R\rangle \langle L|}{\langle L|R\rangle}, \tag{11}$$

where $\mathrm{Tr}_{>t}$ traces over the $T-t$ temporal sites above the cut and leaves the t sites below it open. Thus, $\mathcal{T}_t$ acts on the first t temporal sites. The denominator normalizes the matrix so that $\mathrm{Tr}\,\mathcal{T}_t = 1$. Since $\langle L|$ is not the Hermitian conjugate of $|R\rangle$, $\mathcal{T}_t$ is generally non-Hermitian and may have complex eigenvalues. The consequences of this complex spectrum for truncation are discussed in the next subsection. Figure 3 shows the position of $\mathcal{T}_t$ within the strip geometry.



(a) the strip

(b) the tensor network

(c) $\mathcal{T}_t$

Figure 3: **Position of the reduced transition matrix.** (a) The cut lies on one spatial bond of the strip and extends from the bottom edge to height t. (b) In the tensor-network representation, the horizontal bonds across the cut remain open over the first t temporal sites, while the two sides remain contracted over the remaining $T-t$ sites. (c) Contracting the columns to the left and right of the cut gives $\langle L|$ and $|R\rangle$, respectively. The power method of Section 6 converges these states to the dominant left and right eigenvectors of $\mathcal{E}$. Their contraction over the upper $T-t$ sites gives $\mathcal{T}_t$ in Eq. (11), with the lower t indices left open.

The spectrum of $\mathcal{T}_t$ defines the generalized temporal entropies associated with the transition between the two boundary states. We use the Rényi-2 member of the family introduced in Section 2,

$$S_2^{\mathrm{gen}}(t) = -\log \mathrm{Tr}\left(\mathcal{T}_t^2\right). \tag{12}$$

When $\langle L|$ and $|R\rangle$ differ, there is no gauge in which $\mathcal{T}_t$ is locally diagonal, so its spectrum cannot be obtained from a local decomposition. We choose the Rényi-2 entropy because $\mathrm{Tr}(\mathcal{T}_t^2)$ can instead be evaluated directly by contracting two copies of the RTM, without diagonalizing it. This simplification comes with finite-time corrections that must be accounted for when comparing with the conformal predictions, as discussed in Section 2.2.

Once $\langle L|$ and $|R\rangle$ have converged for a target evolution time T, we evaluate Eq. (12) at every internal bond of the temporal chain, corresponding to cuts $t \in (0, T)$. The entropy is generally complex because $\mathcal{T}_t$ is non-Hermitian, so its real and imaginary parts are analysed separately as functions of the normalized cut position.

The temporal region represented by $\mathcal{T}_t$ extends from the lower edge of the strip to the bond at t and therefore has only one endpoint in the interior. Appendix A.1 explains how this edge-anchored geometry fixes the entropy coefficient; if the interval instead had both endpoints in the interior, the logarithmic slope in Eq. (5) would be twice as large. The central charges reported in Section 7 are extracted using the edge-anchored geometry.

3.4 Truncation based on RDM and RTM

We use two complementary truncation schemes based on the RDM and the RTM. The conventional RDM truncation, available in most tensor-network libraries, brings a boundary MPS into mixed canonical form and discards the smallest singular values of its orthogonality centre, giving the optimal low-rank approximation to that state [6, 30]. The gauge-invariant scheme implemented in `ITransverse.jl` [28] instead truncates the RTM in Eq. (11), which depends on both boundaries. We use the RTM scheme for production calculations and retain the RDM scheme as a complementary check. Appendix B provides further details on both truncations, including their response to gauge transformations and the conditions for optimal truncation. Here we focus on their main differences: their computational cost and the quantities discarded during truncation.

The RTM scheme is substantially cheaper because the two schemes preserve different information. The eigenvalues of the boundary RDM are the squared Schmidt coefficients of $|R\rangle$ across the cut. Truncating them retains the directions in which $|R\rangle$ has the greatest weight, irrespective of their overlap with $\langle L|$. The resulting approximation therefore preserves components of $|R\rangle$ that do not contribute to $\langle L|R\rangle$. By contrast, the RTM weights each direction by its contribution to this overlap: a large component of $|R\rangle$ carries little weight if it is orthogonal to the corresponding component of $\langle L|$. Thus, the RTM can in principle represent the relevant part of both boundaries with fewer retained values and a smaller bond dimension.

The lower cost of the RTM scheme comes with a weaker truncation guarantee. For a Hermitian, positive RDM, the singular values coincide with the eigenvalues, so retaining the largest Schmidt values gives the optimal approximation at a fixed bond dimension. The RTM is instead non-Hermitian, and its eigenvalues may be complex or negative. Although its normalized trace satisfies $\operatorname{Tr}\mathcal{T}_t = \sum_i \lambda_i = 1$, contributions from different eigenvalues may cancel. There is therefore no longer a one-to-one correspondence between the eigenvalues and the singular values. The overlap error can nevertheless be bounded: if the truncation changes the left and right temporal states by at most ϵ_L and ϵ_R in norm, respectively, then the error in their overlap is bounded by $\epsilon_L + \epsilon_R + \epsilon_L\epsilon_R$ [31].

The RTM truncation targets contributions to the overlap rather than minimising these norm errors, so the discarded singular values cannot be used directly to evaluate the bound; however, the scheme works in practice for the physical reasons discussed in Appendix B. When its results are noisy or difficult to interpret, we use the more expensive RDM scheme as a fallback. Appendix F.8 verifies that the two schemes agree at every evolution time where both were run.

4 The ANNNI-type Model

The framework of Sections 2 and 3 requires a quench to a critical point. We study a self-dual quantum spin chain that extends the transverse-field Ising model with NNN interactions [20]. For spin-1/2 degrees of freedom, the Hamiltonian is

$$H = -\sum_i \Big[\sigma_i^z \sigma_{i+1}^z + \lambda\, \sigma_i^x \; + \; p\,(\sigma_i^z \sigma_{i+2}^z + \lambda\, \sigma_i^x \sigma_{i+1}^x)\Big], \tag{13}$$

where $\sigma^{x,z}$ are Pauli matrices, λ is the transverse field, and p controls the strength of the additional interactions. The $\sigma^x\sigma^x$ term preserves self-duality under the Kramers–Wannier transformation [20], which fixes the critical point at $\lambda = 1$. We work at this point throughout, so that the conformal predictions of Section 2 apply.

At $p = 0$, Eq. (13) reduces to the integrable transverse-field Ising chain. For $p > 0$, a Jordan–Wigner (JW) transformation no longer maps the Hamiltonian to free fermions, and the model is therefore non-integrable. The origin of the fermionic interaction is detailed in Appendix C.

4.1 Equilibrium criticality and the central charge

Previous finite-size scaling studies find that the chain remains critical and belongs to the Ising universality class up to $p \lesssim 1.5$, with central charge $c \approx 1/2$ [20]. The universality class of this same model has also been established by a different method, which extracts the conformal data directly from the lattice Hamiltonian [32]. We restrict our simulations to $0 \leq p \leq 1.5$, the range over which the model is known to remain critical and in the Ising universality class [20].

We verify the equilibrium universality class over this range using the ground-state entanglement. For a critical open chain of length L, CFT predicts that the von Neumann entropy of a block of length l scales as [24, 30, 33]

$$S_{\mathrm{vN}}(l, L) = \frac{c}{6} W(l, L) + k, \tag{14}$$

where W is the chord variable of Eq. (4) and k is a non-universal constant. The von Neumann entropy is well suited to this calculation because DMRG provides the full ground-state Schmidt spectrum directly, and also has weaker finite-size corrections than the higher-order Rényi entropies [34].

For each value of p, we obtain the ground state of a chain with $N = 400$ sites and evaluate the entropy across every bond. Equation (14) predicts a linear dependence on $W(l, L)$, so the entanglement profile of a single ground state provides an estimate of the central charge. We fit only the middle part of the chain, using the cuts satisfying $0.25L < l < 0.75L$, where boundary corrections are smallest. The resulting profiles are shown in Figure 4.

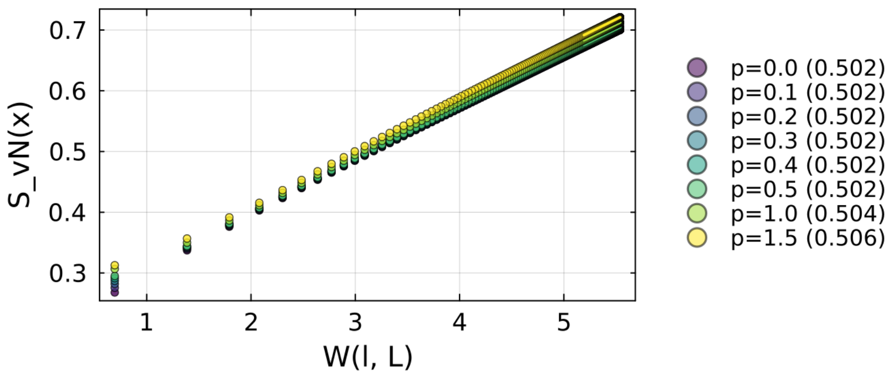


Figure 4: **Effect of NNN interactions to the equilibrium central charge.** Ground-state entanglement profiles at $\lambda = 1$ and $N = 400$, for $p \in [0.0, 1.5]$. The linear dependence on W gives a central charge close to the Ising $c = 1/2$ at every p, with a small common offset from finite-size corrections.

The profiles follow the conformal scaling law at every coupling, and the fitted central charge barely changes with p. Its small common offset from the exact value $c = 1/2$ is consistent with residual finite-size corrections to Eq. (14), since a similar offset is already present at the exactly solvable point $p = 0$. The additional interactions therefore do not change the universality class over the range considered.

The profiles separate at small W, where the cuts lie closest to the open boundaries. Equation (14) is less accurate in this region because the boundaries introduce corrections that depend on the microscopic interactions and therefore on p. This is why these cuts are excluded from the fit. Towards the centre of the chain, these boundary effects weaken and the profiles converge towards the common linear behaviour predicted by CFT. Appendix D.1 gives an independent estimate from the finite-size scaling of the mid-chain entropy and a cross-check using the Rényi-2 entropy.

4.2 Sound velocity

The NNN interaction modifies the propagation of low-energy excitations. At a critical point, the gap closes and the dispersion $\omega(k)$ becomes linear in the lattice momentum k near the gap-closing point. Its slope defines the sound velocity through $\omega(k) = v|k|$, which sets the relativistic scale of the continuum theory. This velocity enters the conformal predictions of Section 2, so it must be determined independently before they can be compared with the dynamical data. For the integrable chain, the exact dispersion $\omega(k) = 2|\sin k|$ gives $v = 2$ [35]; for $p > 0$, we determine v numerically.

We extract $v(p)$ from the equilibrium dispersion of periodic chains, obtained by exact diagonalization, by identifying the lowest excitation with non-zero momentum. Because exact diagonalization is restricted to finite rings, we extrapolate the velocities obtained from five system sizes using their leading $1/N^2$ finite-size dependence. This procedure reproduces the exact result at $p = 0$ and is then applied at each non-zero coupling p.

The velocity grows steadily from 2.000 at $p = 0$ to 10.799 at $p = 1.5$, as shown in Figure 5. We use the independently extrapolated value at each coupling in all subsequent conversions. Although the velocity is non-universal, it is not a fitted parameter in our analysis: it is fixed by the model and determined from the equilibrium spectrum. Appendix D.2 describes the dispersion calculation, extrapolation and validation in detail.

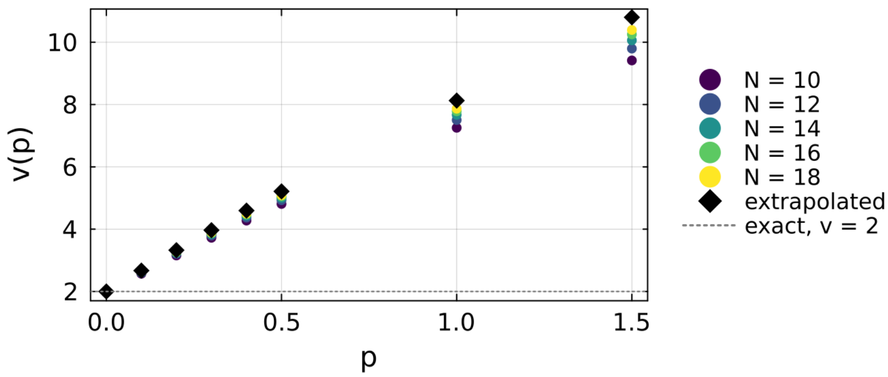


Figure 5: **The sound velocity grows with frustration.** $v(p)$ at $\lambda = 1$, from the equilibrium dispersion of periodic rings of $N = 10$ to 18 sites (coloured circles), together with the values extrapolated to the thermodynamic limit (black diamonds), which are the ones used throughout. The dotted line marks the exact Ising value $v = 2$ at $p = 0$.

5 Building the Time-Evolution MPO

For nearest-neighbour (NN) Hamiltonians, the evolution operator $U(\delta t) = e^{-iH\delta t}$ is usually represented as a Trotter–Suzuki circuit of two-site gates, giving rise to the usual brick-wall structure [13]. However, an interaction between sites i and $i+2$ cannot be assigned to a single bond within this pattern. To preserve the one-site translation invariance required by the transverse contraction, as discussed in Section 3.1, we instead express the Hamiltonian as a finite-state machine (FSM) and exponentiate the resulting representation directly.

5.1 The Hamiltonian as a FSM

In the FSM representation [36, 37], the Hamiltonian is encoded as an MPO whose local tensor has a block-upper-triangular form [38]. Two identity blocks propagate the initial and final states, while the on-site block D generates local terms. The block row C initiates interactions, the block column B terminates them, and the memory block A propagates unfinished interactions between their endpoints. The virtual index therefore records whether an interaction has not yet begun, is being propagated, or has terminated. For more complicated Hamiltonians, this block structure can be generated automatically from the operator [39]. For the ANNNI-type model, the local tensor has bond dimension $D_W = 5$ and takes the form

$$W_H = \begin{pmatrix} \mathbb{1} & C & D \\ 0 & A & B \\ 0 & 0 & \mathbb{1} \end{pmatrix} = \begin{pmatrix} \mathbb{1} & -\sigma^z & -p\lambda\,\sigma^x & -p\,\sigma^z & -\lambda\,\sigma^x \\ 0 & 0 & 0 & 0 & \sigma^z \\ 0 & 0 & 0 & 0 & \sigma^x \\ 0 & \mathbb{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbb{1} \end{pmatrix}. \tag{15}$$

Figure 6 represents Eq. (15) through the allowed paths between the five virtual states. Each node denotes a value of the virtual index, while an arrow from state a to state b, labelled by an operator O, represents the tensor entry $(W_H)_{ab} = O$. The FSM moves from left to right along the chain, beginning in state 1 and ending in state 5. The loop $1 \to 1$ applies identities until a Hamiltonian term is initiated. A direct transition $1 \to 5$ applies the on-site field and completes the term at the same site. An NN interaction instead

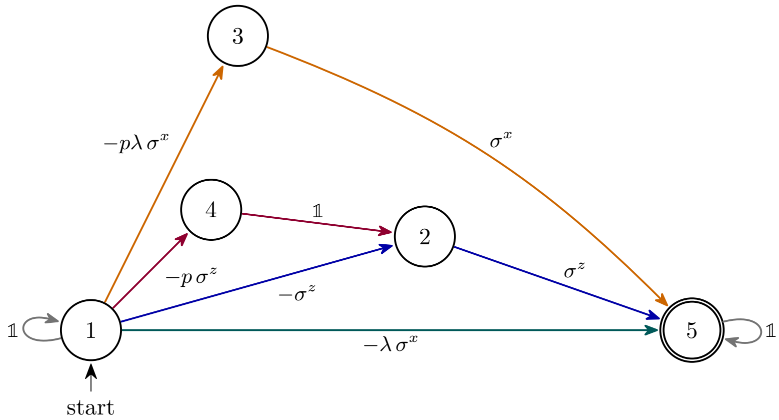


Figure 6: **The Hamiltonian as a finite-state machine (FSM).** The nodes are the five virtual states of Eq. (15); each arrow is labelled by the operator applied during the transition, and every path from state 1 to state 5 (double circle) generates one term of the Hamiltonian. The arrows are coloured according to the generated term: the transverse field in teal, the NN couplings in blue ($\sigma^z\sigma^z$) and orange ($\sigma^x\sigma^x$), and the NNN coupling in purple. The grey loops apply identities before and after the Hamiltonian term. State 2 is shared by the blue and purple paths because both terminate with σ^z, keeping the bond dimension at $D_W = 5$.

follows either $1 \to 2 \to 5$ or $1 \to 3 \to 5$: the first transition applies the operator at site i and records whether a σ^z or σ^x endpoint remains to be applied; the second applies the corresponding operator at site $i+1$ and completes the interaction. The NNN term follows $1 \to 4 \to 2 \to 5$: the first transition applies σ^z at site i, the second inserts an identity at site $i+1$, and the third applies σ^z at site $i+2$. After state 5 is reached, the loop $5 \to 5$ applies identities along the remainder of the chain. Zero entries in W_H correspond to absent arrows, so no other transitions are allowed.

Because the FSM is directional, its local tensor is asymmetric under interchange of the virtual indices. The exponentiated operator inherits this structure, as does the spatial transfer matrix after moving to the transverse picture. The resulting operator $\mathcal{E}$ is generally non-Hermitian and non-normal, $\mathcal{E}\mathcal{E}^\dagger \neq \mathcal{E}^\dagger\mathcal{E}$. Its left and right eigenvectors are therefore not related by Hermitian conjugation, and we compute the two temporal boundary states separately.

5.2 Exponentiating the FSM

We seek an approximation $W(\tau) \approx e^{\tau H}$, with $\tau = -i\delta t$, constructed from the blocks of W_H. Neither of the most direct approaches gives a suitable many-body operator: exponentiating the local tensor as an ordinary matrix mixes its operator-valued blocks but does not reproduce the exponential of the full Hamiltonian; on the other hand, a finite-order Taylor expansion of $e^{\tau H}$ avoids this local exponentiation, but it is not size-extensive because H^n contains $\mathcal{O}(N^n)$ contributions [40].

We instead use a construction that organises the expansion according to the spatial overlap of the local Hamiltonian terms [40]. Contributions with disjoint support commute and factorise, allowing them to be retained to all orders. Only connected clusters of overlapping terms must be reproduced explicitly. We use the second-order member of this family, denoted *VD2*, whose virtual bond dimension is 13 for the model of Eq. (13). Appendix E gives the full construction, the explicit tensor and the origin of this bond dimension.

Cheaper implementations—with lower bond dimension—include the operators W^{I} and W^{II} [41], which keep only part of the contributions from overlapping Hamiltonian terms. Both are first order in a single application, for this model and for a strictly NN chain alike, and reaching second order with them means composing several stages into one time step. Appendix E.2 measures how the error of the three operators falls as δt is refined. We therefore use *VD2*, which is second order in one application, despite its larger bond dimension.

After moving to the transverse picture, the virtual bond dimension of the evolution MPO becomes the local physical dimension of the temporal chain. The *VD2* construction therefore has a larger local dimension than W^{II}. Its second-order accuracy permits the larger time step $\delta t = 0.1$ for a given discretisation error, which reduces both the temporal length at fixed physical time and the number of required truncation steps, making the long-time calculations of Section 7 more computationally accessible with *VD2*.

6 The Block Power Method

Section 3 reduced the dynamical calculation to finding the leading eigenpairs of the transfer matrix $\mathcal{E}$. We obtain them with a block power method that evolves several temporal states simultaneously. The method returns the dominant eigenvalues and their left and right eigenvectors, which we then identify and order for comparison with the CFT predictions. Appendix F gives the implementation details.

The right and left eigenvectors of $\mathcal{E}$ satisfy

$$\mathcal{E} \left| R_i \right\rangle = \mu_i \left| R_i \right\rangle , \qquad \left\langle L_i \right| \mathcal{E} = \mu_i \left\langle L_i \right| . \tag{16}$$

Because $\mathcal{E}$ is non-normal (Section 5.1), its left and right eigenvectors differ, and neither set is orthogonal by itself. Instead, the two sets are biorthogonal, $\langle L_i | R_j \rangle \propto \delta_{ij}$. Here the bra notation does not imply complex conjugation: $\langle L_i|$ is a row vector, and $\langle L_i | R_j \rangle$ denotes the direct contraction of the two temporal networks. We therefore propagate the left states with the transpose $\mathcal{E}^{\mathsf{T}}$, which implements the second equation in Eq. (16).

6.1 The algorithm

The single-vector power method implemented in `ITransverse.jl` [28] computes only the dominant eigenpair. This is sufficient for computing the Loschmidt rate from the modulus of the dominant eigenvalue μ_0, the central charge from its phase, and the left and right dominant states ($\langle L_0| , |R_0\rangle$) required for the generalized temporal entropies. It cannot, however, determine the boundary operator content, which is encoded in the phase gaps between μ_0 and the subleading eigenvalues. Accessing these gaps requires the individual subleading eigenvalues and therefore a block power method. Their resolution becomes increasingly demanding with evolution time: from Eq. (7), the i-th phase gap scales as $\pi x_i/(vT)$, decreasing as $1/T$ at fixed coupling and becoming smaller still as the velocity increases with p.

We resolve these closely spaced eigenvalues by iterating a block of k states, which selects the leading subspace collectively. Ordering the eigenvalues by decreasing modulus, after m applications of $\mathcal{E}$ the component along the first discarded eigenvector is suppressed relative to the last retained one by $(|\mu_k|/|\mu_{k-1}|)^m$. The convergence rate is therefore controlled by the separation between the retained subspace and the rest of the spectrum, rather than by gaps between eigenvalues within the block. Once this subspace has converged, a projected eigenproblem separates its individual eigenpairs.

To represent the two subspaces, the block power method evolves k right states $|R_j\rangle$ and k left states $\langle L_j|$, each represented as a temporal Matrix Product State (tMPS). We initialise the two blocks independently with random complex tensors. Algorithm 1 summarises one iteration, and Figure 7 shows the tensor-network operations used to apply and project the transfer matrix.

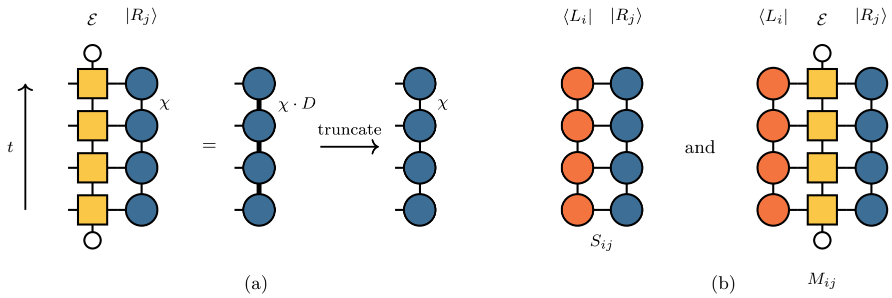


Figure 7: **One iteration of the block power method.** Each tMPS forms a column, while $\mathcal{E}$ is the corresponding column of the space–time network closed at both ends by the product state (small open circles). (a) Applying $\mathcal{E}$ to one member of the right block, as in Eq. (17), increases the temporal bond dimension from χ to χD, where D is the tMPO bond dimension. The result is then truncated back to χ. The left block is treated analogously using the transpose. (b) Contractions defining the two $k \times k$ matrices in Eq. (19). The left chain enters directly as a bra, without complex conjugation.

Algorithm 1 Block power method for the k dominant eigenpairs of $\mathcal{E}$.

1: $|R_j\rangle$, $\langle L_j| \leftarrow$ random complex tMPSs, $\quad j = 1, \dots, k$
2: **repeat**
3: $\quad \left|\tilde{R}_j\right\rangle \leftarrow \text{truncate}_\chi\left(\mathcal{E}\,|R_j\rangle\right), \qquad \left\langle\tilde{L}_j\right| \leftarrow \text{truncate}_\chi\left(\langle L_j|\,\mathcal{E}\right)$ ▷ apply
4: $\quad S_{ij} \leftarrow \langle L_i|R_j\rangle, \qquad M_{ij} \leftarrow \left\langle L_i\middle|\tilde{R}_j\right\rangle = \langle L_i|\,\mathcal{E}\,|R_j\rangle$ ▷ project
5: $\quad$ solve $Mv = \theta\, Sv$ for Ritz values $\theta_1, \dots, \theta_k$ and coefficients V, U
6: $\quad |R_j\rangle \leftarrow \text{truncate}_\chi\left(\sum_i V_{ij}\left|\tilde{R}_i\right\rangle\right), \qquad \langle L_j| \leftarrow \text{truncate}_\chi\left(\sum_i U_{ij}\left\langle\tilde{L}_i\right|\right)$ ▷ de-mix
7: $\quad$ normalise every block member
8: **until** the leading Ritz values stop changing within tolerance

The first step applies the transfer matrix to each member of the right block and its transpose to each member of the left block,

$$\left|\tilde{R}_j\right\rangle = \mathcal{E}\,|R_j\rangle\,, \qquad \left\langle\tilde{L}_j\right| = \langle L_j|\,\mathcal{E} \iff \left|\tilde{L}_j\right\rangle = \mathcal{E}^{\mathsf{T}}\,|L_j\rangle\,. \tag{17}$$

Applying the tMPO multiplies the bond dimension of each state by D, as shown in Figure 7a. After each application, we truncate the propagated state back to the maximum bond dimension χ using either scheme from Section 3.4 and normalise it, keeping the computational cost of each iteration constant.

The next step projects the eigenproblem onto the current block subspaces. We approximate a right eigenvector by

$$|\psi\rangle = \sum_{j=1}^{k} v_j\,|R_j\rangle\,. \tag{18}$$

The ansatz is a combination of the current block members and not an exact eigenvector, so inserting it into $\mathcal{E}\,|\psi\rangle = \theta\,|\psi\rangle$ leaves a residual. We require its contraction with every member of the left block to vanish, so that $\langle L_i| (\mathcal{E} - \theta) |\psi\rangle = 0$. This condition reduces the full eigenproblem to the two $k \times k$ matrices shown in Figure 7b,

$$S_{ij} = \langle L_i|R_j\rangle\,, \qquad M_{ij} = \langle L_i|\,\mathcal{E}\,|R_j\rangle\,, \tag{19}$$

and the generalized eigenvalue problem

$$\sum_j M_{ij}\, v_j = \theta \sum_j S_{ij}\, v_j. \tag{20}$$

The overlap matrix S describes the relative orientation of the left and right subspaces, while M is the transfer matrix projected between them. Solving Eq. (20) yields approximations θ_j to $\mu_0, \ldots, \mu_{k-1}$, which we order by modulus. It also yields the coefficient matrices V and U used in Algorithm 1; the columns of V specify the combinations of propagated right states associated with the approximate right eigenvectors, and U provides the corresponding combinations for the left block. If S is ill-conditioned, the oblique projection amplifies numerical errors, so we use the pseudoinverse regularisation described in Appendix F when solving Eq. (20).

We then use the Ritz coefficient matrices to separate the propagated blocks into approximate eigenpairs. Before the next iteration, we truncate the resulting states back to χ and normalise them. The iteration stops once the leading Ritz values remain stable within the prescribed tolerance.

The tensor-network operations dominate the computational cost. Most of the cost arises from the sums used to separate the propagated block states. Appendix F.4 quantifies these contributions and describes the measures used to reduce them.

When only the eigenvalues are required, we retain the blocks as orthonormal bases of the leading invariant subspaces instead of assigning individual members to separate eigenvectors. This representation is better conditioned near degeneracies and does not change the Ritz values, as discussed in Appendix F.2. We use it for the late-time spectral measurements in Section 7.

6.2 Selecting and tracking the physical branch

From the k Ritz values returned at each evolution time, we identify the dominant eigenvalue μ_0 and order the remaining tower states. At short times, where the leading spectral gap is wide, μ_0 is unambiguously selected by largest modulus. At later times, emergent dual unitarity drives the tower eigenvalues towards a common modulus (Section 2.2); if the two leading candidates differ in modulus by less than one per cent, we select the one closest in the complex plane to the previously accepted μ_0, thereby enforcing phase continuity. The subleading eigenvalues are then ordered by increasing phase distance from μ_0, matching the structure of the conformal tower.

When computing a ladder of evolution times, we initialise the iteration at $T + \Delta T$ with the converged block from T. This continuation tracks the physical branch smoothly from one time step to the next. If the block contains too few states to resolve the required operator content, we expand the block in stages ($k = 4 \to 6 \to 8$), initialising each larger subspace from the converged smaller one.

Convergence alone does not guarantee that the iteration reaches the physical fixed point; since the blocks are randomly initialised, different seeds can converge to different, non-physical states at long evolution times. Repeating the calculation from independent seeds shows that these failures are rare at short times and increasingly frequent as T grows, and that they affect the eigenvectors far more than the eigenvalues: $|\mu_0|$ reproduces across seeds at every usable time, while the entropies built from the eigenvectors do not (Appendix F.8). To filter out these artefacts without biasing the results towards CFT predictions, we apply two consistency checks. First, we require $|\mu_0|$ to vary continuously with T, recomputing any discontinuous point with a fresh random seed. Second, when extracting eigenvectors for the entropy, we repeat the calculation until two independent runs agree. Since two random seeds can occasionally converge to the same unphysical fixed point, reproducibility alone is insufficient; we therefore also require the accepted entropy to vary smoothly with neighbouring evolution times.

We determine the upper limit of each time window from the winding of the leading eigenvalue. Along the physical branch, the phase of μ_0 advances at an approximately constant rate with T, so any persistent departure from this rate signals that the branch is no longer resolved, and therefore sets the boundary of the usable fitting window. Appendices F.6 and F.7 give the corresponding selection safeguards and conditioning diagnostics.

7 Results

We now test the CFT predictions of Section 2.2 against simulations of the critical ANNNI-type chain (Eq. (13) with $\lambda = 1$), quenched from the polarized state $|X+\rangle$. We vary the NNN coupling from the integrable point, $p = 0$, up to $p = 0.5$. Because extracting the universal data reliably requires a sufficiently wide time window, and the simulation cost grows significantly with p, we restrict the dynamical calculations to $p \leq 0.5$ for simplicity and because it suffices for extracting the universal properties from the critical system.

The analysis divides into two complementary routes. The entropy route uses the transfer-matrix eigenvectors to compute the generalized Rényi-2 entropy (Eq. (12)) and compares it with the conformal profile (Eq. (5)). The spectral route uses the eigenvalues alone to verify emergent dual unitarity, extract the central charge from the phase curvature of the leading eigenvalue (Eq. (6)), and determine the boundary exponent from the first spectral gap (Eq. (7)). In both routes, we evaluate the integrable Ising point ($p = 0$) alongside the non-integrable cases ($p > 0$) to test whether the conformal predictions remain valid in the presence of the NNN interaction. Table 1 collects the universal data obtained from both routes, and the rest of this section explains how each value is measured.

Unless stated otherwise, all simulations use the second-order *VD2* MPO of Section 5 with time step $\delta t = 0.1$, cooling regulator $\beta_0 = 0.2$ ($N_\beta = 4$), and maximum temporal bond dimension $\chi = 64$. We convert all spectral quantities into universal data using the independently determined sound velocities $v_\infty(p)$ from Section 4.2, with fitting conventions and controls detailed in Appendix G.3.

p	v_∞	window	spectral c	x_1	entropy-route c
0	2.000	$T \leq 20$	0.45	0.498	[OPEN]
0.1	2.670	$T \leq 9$	0.47	0.497	[OPEN]
0.3	3.967	$T \leq 6.5$	0.49	0.493	[OPEN]
0.5	5.212	$T \leq 5.5$	0.53	0.487	[OPEN]

Table 1: **Universal data against the NNN coupling.** The spectral central charge comes from Eq. (6) on ladders sampled at $\Delta T \leq 0.5$ inside the window, with the branch constant pinned. The boundary exponent is the first phase gap converted through Eq. (7). Appendix G.3 gives the fit windows and the controls behind these numbers. The entropy-route central charge is the extrapolated plateau of Section 7.1.

7.1 The entropy route

At the integrable point, the generalized temporal entropy follows the conformal prediction established for the Loschmidt echo of the critical Ising chain [17]. We find that this behaviour persists when the NNN interaction is introduced. Figure 8 shows the Rényi-2 profiles at $p = 0.1$. The real part follows the chord dependence in Eq. (5), while the imaginary part is approximately constant in the interior of the strip. For this comparison, we fix $c = \frac{1}{2}$ and adjust only the non-universal constant s_0, following the analysis of the integrable chain. Appendix G.2 presents the corresponding profiles at the remaining couplings.

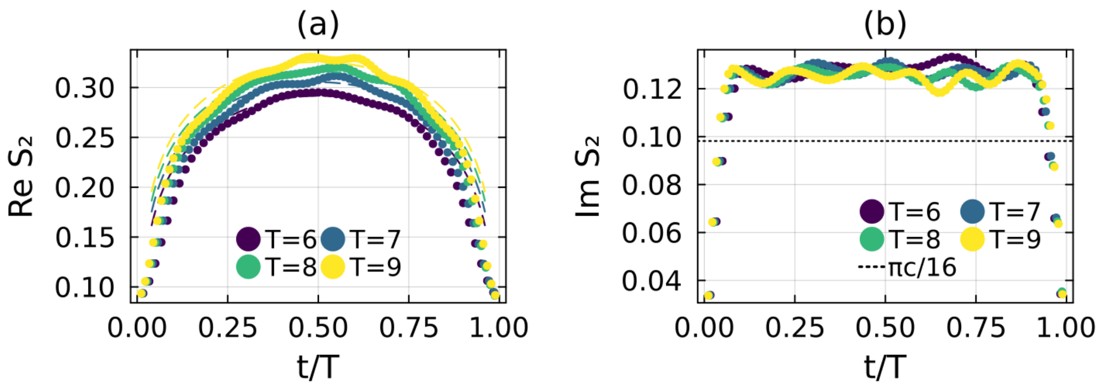


Figure 8: **The generalized temporal entropy at $p = 0.1$.** Each colour denotes one evolution time. (a) Real part as a function of t/T. The dashed curves show the conformal profile in Eq. (5) with $c = \frac{1}{2}$. (b) Imaginary part, which is approximately constant in the interior of the strip. Its plateau lies above the asymptotic value $\pi c/16$ (dashed line) because of finite-time corrections that decrease with increasing T.

At every simulated time, the measured plateau lies above the asymptotic value $\pi c/16$. This offset is the finite-time correction discussed in Section 2.2 [19]. Because it decays slowly with T, the asymptotic plateau must be obtained by extrapolating a sequence of evolution times. A short time series cannot distinguish the decay from a constant offset and therefore does not constrain the asymptote. Including the finite-time correction gives $c =$ [OPEN] at the integrable point, consistent with the exact value $\frac{1}{2}$, and $c =$ [OPEN] at $p = 0.1$. In contrast to Figure 8, where c is fixed to test the predicted profile, these extrapolations determine c from the data. The results are included in Table 1.

We use the imaginary plateau rather than the real-part chord slope to estimate the central charge. At each evolution time, Eq. (5) predicts the plateau value directly. By contrast, extracting c from the real part requires a two-parameter fit of the chord coefficient and the non-universal offset s_0, making the result more sensitive to the interval of the strip included in the fit.

The accessible time window decreases as p increases, leaving fewer points with which to determine the finite-time decay. The latest accessible times also require the repetition procedure described in Section 6.2. The repetition also measures where the route ends: the fraction of independent runs returning an unphysical profile grows with T at every coupling, from the first failures at $T = 18$ at the integrable point to every run at $p = 0.5$, $T = 6$ (Appendix F.8). When the available window is too short to constrain the extrapolation, we do not report an entropy-based central charge and instead use the spectral method below.

7.2 The spectral route

Using the eigenvalue-only method of Section 6.1, we compute the spectrum up to $T = 20$ at the integrable point and up to the end of each window in Table 1 at the remaining couplings. Only the shorter fitting windows listed in Table 1 are used to extract universal quantities. Figure 9 shows the trajectory of μ_0 in the complex plane after normalising each time series by its mean modulus. The phase winds throughout the simulated interval, while the modulus varies by less than one per cent. This behaviour is consistent with the emergent dual unitarity described in Section 2.2. The corresponding central-charge fits are shown in Figure 10.

We extract the central charge from the phase curvature using Eq. (6). Each time series is sampled with $\Delta T = 0.5$ because the fit is constrained by both the extent of the window and the number of points within it. The fitting conventions are given in Appendix G.3.

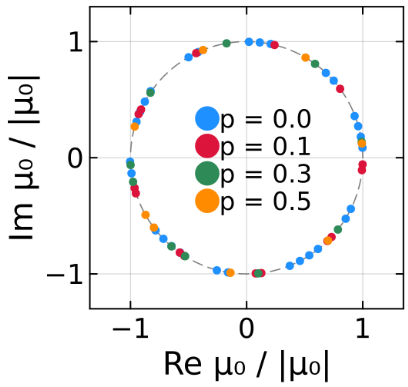


Figure 9: **Trajectory of the normalised leading eigenvalue.** At each coupling, μ_0 is divided by its mean modulus over the time series. The phase winds with T, while the weak variation of the modulus keeps the points close to the dashed unit circle.

Table 1 summarises the results. The spectral estimates remain within approximately ten per cent of the Ising value at every coupling and are consistent with $c = \frac{1}{2}$ within the resolution of the fits. The equilibrium chord analysis of Section 4.1 gives $c \approx 0.502$ over the same range. These determinations use independent data: the equilibrium estimate is obtained from the ground state, whereas the spectral estimate follows from the real-time dynamics.

The sound velocity has opposite effects on finite-time corrections and numerical resolution. Since the corrections to the boundary dimensions are controlled by vT, a larger v brings the measured dimensions closer to their conformal values at fixed T. At the same time, the phase gaps decrease as $1/(vT)$ and become harder to resolve, while converting the fitted phase curvature into a central charge amplifies its numerical uncertainty by a factor of v. The fitting windows in Table 1 therefore shorten as p increases, despite the smaller finite-time corrections within those windows.

The Ising boundary tower persists when the NNN interaction is introduced. As shown in Figure 11 and Table 1, the leading dimension remains close to $\frac{1}{2}$ at every coupling. At fixed evolution time, its deviation from this value decreases as p increases, consistent with finite-time corrections controlled by vT.

Resolving the higher levels requires a larger block. With $k = 8$, the first three dimensions approach the Ising values $\frac{1}{2}$, $\frac{3}{2}$, and 2 at every non-integrable coupling shown in Figure 11. The remaining Ritz values do not converge sufficiently within the accessible time windows, so we do not assign them to higher members of the conformal tower.

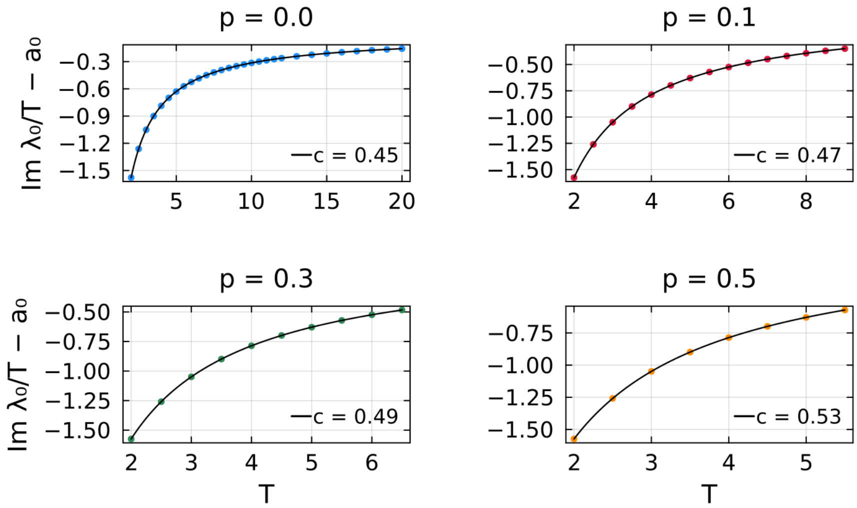


Figure 10: **Central charge from the phase of the leading eigenvalue.** Fits of Eq. (6) to the phase of μ_0, shown separately for each coupling. The fitted constant is subtracted to display the time-dependent contribution. The curvature determines the central charge reported in each legend.

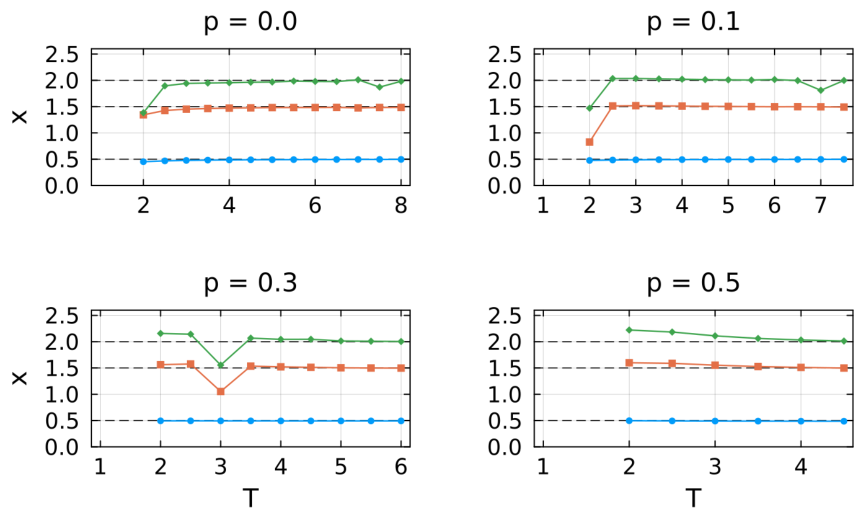


Figure 11: **Boundary dimensions as a function of evolution time.** The three leading dimensions at each coupling are obtained from the phase gaps through Eq. (7). The series show x_1 (circles), x_2 (squares), and x_3 (diamonds), in ascending order; dashed lines mark the Ising values $\frac{1}{2}$, $\frac{3}{2}$, and 2. The leading dimension remains close to $\frac{1}{2}$, while the higher dimensions approach their predicted values as T increases. Every coupling uses $k = 8$.

8 Conclusions and Outlook

This thesis asked whether the conformal signatures obtained by transverse contraction persist when integrability is broken. For the self-dual ANNNI-type chain and the couplings studied here, the resolved signatures remain consistent with the Ising universality class. The dominant transfer-matrix eigenvalue retains an approximately time-independent modulus and a winding phase, as predicted by emergent dual unitarity. The curvature of this phase gives a central charge compatible with $c = \frac{1}{2}$ within the resolution of the fits, while the leading phase gaps reproduce the Ising boundary dimensions (Table 1). Within the accessible critical regime, the NNN interaction therefore changes the non-universal scales without altering the universal content that we can resolve.

Extending the calculation to longer-range interactions required a corresponding transfer-matrix construction. We represent the Hamiltonian as a FSM, exponentiate it, and rotate the resulting MPO into a column of the space–time network. The standard constructions of this operator are first order in a single application. We therefore use the *VD2* construction, which reproduces every pair of overlapping Hamiltonian terms exactly and is second order in one application, at the cost of a larger virtual bond. We determine the leading spectrum with a block power method that propagates the left and right temporal states together while preserving their biorthogonal pairing. When only the spectrum is required, an eigenvalue-only variant propagates well-conditioned bases of the leading subspaces rather than resolving the individual eigenvectors.

The equilibrium calculations supply the non-universal input and provide an independent reference for the dynamical results. The DMRG estimates of the central charge remain consistent with the Ising value up to $p = 1.5$, while exact diagonalisation of finite rings determines the sound velocity at each coupling (Section 4). This independently measured velocity converts the transfer-matrix phase gaps into scaling dimensions. The equilibrium and dynamical estimates of the central charge agree within the resolution of the dynamical fits, although the latter differ from $\frac{1}{2}$ by up to approximately ten per cent over the reported range.

The entropy and spectral analyses reach different evolution times because they require different parts of the eigensystem. First-order perturbation theory shows that eigenvector mixing is inversely proportional to the separation between neighbouring eigenvalues, whereas the corresponding eigenvalue shift contains no spectral-gap denominator. The entropy calculation requires the dominant left and right eigenvectors and therefore loses reliability earlier. By contrast, the spectral calculation requires only the Ritz values of the leading subspace and remains usable over a wider range. Failures also depend on the random initialisation: their frequency increases with evolution time, from the first failures at $T = 18$ for the integrable point to failure of every run at $p = 0.5$ and $T = 6$. The dominant modulus $|\mu_0|$ nevertheless remains reproducible across seeds. Exact diagonalisation at short times indicates that the coupling affects the conditioning of the eigenvector basis more strongly than the spectral gap, but the mechanism responsible for this dependence remains unclear.

The perturbative argument explains why the eigenvalues are more stable than the eigenvectors, but it does not determine when the entropy calculation fails. The phase rigidity decreases with evolution time and more rapidly at stronger coupling, yet its value at the last reliable time varies by four orders of magnitude across the couplings studied. The numerical controls do not identify a single limiting parameter. Increasing the bond dimension does not extend the reliable range, a tighter truncation cutoff changes the termination of the iteration without changing the result, and changing the truncation scheme reproduces the values and lets the iteration reach its tolerance. The reported time windows therefore characterise the reach of the present numerical configuration rather than a fundamental limit of transverse contraction.

The sound velocity provides an independent indication of this numerical reach. Boundary dimensions are obtained from phase gaps proportional to $1/(vT)$, so a larger velocity compresses the spectrum and requires greater phase resolution at fixed evolution time. Because v can be measured in equilibrium, this compression can be estimated before the dynamical calculation.

These limitations suggest several improvements. The stopping criterion should account for the truncation noise floor, since the formal tolerance is not reached under the production truncation even after the result has stabilised. A separate theoretical problem is to determine how the coupling controls the conditioning of the eigenvector basis. Most of the computational cost arises from separating and compressing the propagated block states, so reducing this cost would extend the accessible evolution times. These improvements would allow the dynamical central charge $c(p)$ and leading boundary dimension $x_1(p)$ to be followed towards the boundary of the critical phase, where the equilibrium results remain consistent with Ising criticality but the present dynamical windows are too short for reliable fits.

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A Conformal-field-theory derivations

Section 2 relates the Loschmidt echo of a critical quench to the partition function of a CFT on a strip whose width is determined by the evolution time. The analysis uses two results derived here: the logarithmic coefficient of the generalized temporal entropy and the spectrum of the transfer matrix along the strip. We first obtain the entropy coefficient and chord variable from twist-operator correlation functions, including the factor of two associated with an edge-anchored bipartition. We then quantise the theory on the strip and continue its width to complex time, obtaining Eqs. (2), (6), and (7).

A.1 Conformal maps, the two-point function and the entropy coefficients

In two dimensions, we write the Euclidean coordinates as $z = x + i\tau$ and $\bar{z} = x - i\tau$. A conformal map is a transformation $z \to w(z)$ depending on z alone, together with an independent transformation $\bar{z} \to \bar{w}(\bar{z})$ of the conjugate coordinate. The holomorphic and antiholomorphic dependences therefore transform separately, and we call each one a chiral sector; in real time they carry the left- and right-moving excitations. A primary operator ϕ is one that transforms under such a map with the Jacobian alone, without the derivative terms a general operator acquires, and its conformal weights $(h, \bar{h})$ set the exponent contributed by each sector. Their sum,

$$x = h + \bar{h} \tag{A.1}$$

is the scaling dimension used in the main text, and their difference $s = h - \bar{h}$ is the spin, which measures the response to a rotation about the operator. The twist operators used below sit at branch points of the replica surface, which are invariant under such a rotation, so $h = \bar{h}$ and $s = 0$. On the plane, the two-point function has the power-law form

$$\langle \phi(z_1)\phi(z_2) \rangle = \frac{1}{(z_1 - z_2)^{2h} \, (\bar{z}_1 - \bar{z}_2)^{2\bar{h}}}. \tag{A.2}$$

Under a conformal map $z \to w$, a primary field transforms with the corresponding Jacobian. Equation (A.2) therefore determines its correlator on any geometry conformally related to the plane [23]. The logarithmic map $w = (\beta/\pi) \log z$ maps the plane to a cylinder of circumference β and the upper half-plane to a strip of width β. Here β is a length, so on the lattice it carries the sound velocity through $\beta = v(2\beta_0 + iT)$, and v does not appear separately in the map. The transformed correlator is [17, 24]

$$\langle \phi(\omega_1)\phi(\omega_2) \rangle \;\propto\; \left[\sinh \tfrac{\pi(\omega_1 - \omega_2)}{\beta}\right]^{-2h} \left[\sinh \tfrac{\pi(\bar{\omega}_1 - \bar{\omega}_2)}{\beta}\right]^{-2\bar{h}}. \tag{A.3}$$

The separation on the plane is replaced by the chord distance $(\beta/\pi) \sinh[\pi(\omega_1 - \omega_2)/\beta]$ on the cylinder or strip. Analytic continuation of the width to real time replaces the hyperbolic sine by a sine. This continued chord distance gives the variable W in Eq. (4).

The entropy coefficients follow from the same correlation function. The Rényi entropy of a region $\mathcal{A}$ is

$$S_n = \frac{1}{1-n} \log \operatorname{Tr} \rho_{\mathcal{A}}^n, \tag{A.4}$$

so the quantity to be computed is the moment $\operatorname{Tr} \rho_{\mathcal{A}}^n$, the trace of the nth power of the reduced density matrix. For integer n this moment is itself a partition function: each factor of $\rho_{\mathcal{A}}$ is written as a path integral, and taking the trace joins n copies of the system cyclically along $\mathcal{A}$ [24, 33]. Analytic continuation to $n \to 1$ gives the von Neumann entropy. Each endpoint of $\mathcal{A}$ in the interior is a branch point of the replicated surface and is represented by a primary twist operator of the n-fold replicated theory with scaling dimension [24]

$$x_n = \frac{c}{12}\Big(n - \frac{1}{n}\Big), \tag{A.5}$$

whose weights are $h_n = \bar{h}_n = x_n/2$ in the notation of Eq. (A.1).

For an interval of length ℓ with both endpoints in the interior, the moments are determined by the two-point function of two twist operators. Equation (A.2) gives $\operatorname{Tr}\rho_{\mathcal{A}}^n \sim \ell^{-2x_n}$ and therefore

$$S_n = \frac{c}{6}\Big(1 + \frac{1}{n}\Big)\log\ell + \text{const}, \qquad S_{n\to 1} = \frac{c}{3}\log\ell + \text{const}. \tag{A.6}$$

The temporal bipartitions used in this thesis terminate at an edge of the strip and have only one endpoint in the interior. Their moments therefore involve the one-point function of a single twist operator. A conformal boundary relates the two chiral sectors, so the antiholomorphic part of an operator can be replaced by a holomorphic one at its mirror image [42]. The one-point function is then evaluated as a correlator between the operator and that image, separated by twice the distance to the boundary [24]. Because a single operator is inserted rather than two, the exponent is x_n instead of $2x_n$ and the logarithmic coefficient is halved:

$$S_n = \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\log\ell + \text{const}. \tag{A.7}$$

On a strip of finite extent ℓ, the distance between an operator at x and its image is $(2\ell/\pi)\sin(\pi x/\ell)$, which is exactly the argument of the chord variable $W(x,\ell)$ in Eq. (4), so the edge-anchored Rényi entropy is

$$S_n = \frac{c}{12}\Big(1 + \frac{1}{n}\Big)W(x,\ell) + s_0, \tag{A.8}$$

where s_0 is non-universal. For an open chain the same result is written with ℓ/π rather than $2\ell/\pi$ [24]. This difference contributes only an additive constant and is absorbed into s_0.

To obtain Eq. (3), we continue the strip extent to real time by setting $\ell \to i\,vT$ and $x \to i\,vt$. The ratio x/ℓ remains real, while the prefactor inside the logarithm acquires a factor of i. Using $\log i = i\pi/2$ gives

$$S_n^{\text{gen}}(t) = s_0 + \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\Big[W(t,T) + \frac{i\pi}{2}\Big], \tag{A.9}$$

This reproduces Eq. (3). The real part contains the chord profile, while the imaginary part is the constant $\frac{\pi c}{24}(1 + \frac{1}{n})$, equal to $\pi c/16$ at $n = 2$. The velocity contributes only an additional constant inside the logarithm and is absorbed into s_0.

A.2 The transfer-matrix spectrum from the stress tensor

The stress-energy tensor determines the response of the theory to coordinate transformations. In two dimensions, its modes L_n and $\bar{L}_n$ generate conformal transformations in the two chiral sectors and satisfy one copy of the Virasoro algebra each, whose central term is proportional to c [23]. The mode L_0 generates scale transformations, and its eigenvalues on primary operators are their conformal weights. Its spectrum therefore encodes the operator content.

Conformal boundary conditions at the two edges of the strip identify the chiral sectors. Extending the stress tensor into the lower half-plane through $\bar{T}(z) = T(z)$ leaves a single Virasoro algebra [42]. The dimensions below are therefore the weights h of boundary operators, rather than the bulk dimensions $h + \bar{h}$ defined in Eq. (A.1).

The generator of translations along the strip is obtained from the stress tensor. Since the stress tensor is not a primary field, a conformal map $z \to w$ introduces an inhomogeneous Schwarzian term proportional to the central charge [23]:

$$T(w) = \Big(\frac{\mathrm{d}z}{\mathrm{d}w}\Big)^2 T(z) + \frac{c}{12}\,\{z; w\}. \tag{A.10}$$

The map $w = (\beta/\pi)\log z$ from Appendix A.1 takes the upper half-plane to a strip of width β and has Schwarzian derivative $\{z; w\} = -\frac{1}{2}(\pi/\beta)^2$. Since $\langle T\rangle$ vanishes on the half-plane, its expectation value on the strip is $\langle T(w)\rangle = -\pi^2 c/(24\beta^2)$. Integrating across the strip gives a universal contribution $-\pi c/(24\beta)$ to the free energy per unit length [43]. Including the excitation levels generated by dilatations, the translation generator is [44]

$$\hat{H} = \frac{\pi}{\beta}\Big(L_0 - \frac{c}{24}\Big). \tag{A.11}$$

It contains both universal quantities of interest: the Casimir term $-c/24$ determines the central-charge contribution, while the spectrum of L_0 gives the boundary scaling dimensions. The transfer matrix advances the network by one spatial column, and $\hat{H}$ generates translations in that direction. Hence $\mathcal{E} = e^{-a\hat{H}}$, where a is the column spacing [17]. Substituting Eq. (A.11) and setting $a = 1$ in lattice units reproduces Eq. (2), with $g = a/\beta$ and eigenvalues

$$\mu_i = \exp\Big[-\pi g\Big(x_i - \frac{c}{24}\Big)\Big]. \tag{A.12}$$

The identity sector has $x_0 = 0$ and gives the dominant eigenvalue. Differences between logarithmic eigenvalues therefore cancel the Casimir contribution and isolate the boundary dimensions.

The continuum expression must be supplemented by two non-universal lattice contributions. The free energy per unit length contains a bulk term and a contribution from the two boundaries [43], giving the level energies

$$E_i = f\beta + f_s + \frac{\pi}{\beta}\Big(x_i - \frac{c}{24}\Big), \tag{A.13}$$

where f and f_s depend on the microscopic model, Trotter step, and regulator. Since both terms are common to every level, they multiply all eigenvalues of $\mathcal{E}$ by the same factor. They therefore cancel in the phase differences used to extract the dimensions. The leading phase is instead fitted as a series in T. Since β is proportional to T, the bulk term $f\beta$ contributes a term linear in T, while f_s is real and shifts only the modulus. Neither reaches the $1/T$ coefficient, which is the one carrying the central charge.

The effective strip width contains both imaginary- and real-time contributions. The regulator supplies imaginary-time intervals at the two boundaries, the quench supplies the intervening real-time interval, and the velocity converts these durations into lengths:

$$\beta = v\,(2\beta_0 + i\,T), \qquad \frac{1}{\beta} \;\simeq\; \frac{1}{v}\Big(\frac{2\beta_0}{T^2} - \frac{i}{T}\Big), \tag{A.14}$$

where the second expression is expanded to first order in β_0/T. Substitution into Eq. (2) gives

$$\log\mu_i \;\simeq\; \frac{i\pi}{v\,T}\Big(x_i - \frac{c}{24}\Big) \;-\; \frac{2\pi\beta_0}{v\,T^2}\Big(x_i - \frac{c}{24}\Big). \tag{A.15}$$

The universal contributions appear in the phases at order $1/T$, whereas the regulator changes the moduli at order β_0/T^2. The eigenvalues follow from the level energies as $\log \mu_i = -E_i$ at $a = 1$. Restoring the non-universal terms of Eq. (A.13), setting $x_0 = 0$ for the identity, and taking the imaginary part of $\lambda_0 = \log(-\mu_0)$ gives

$$\operatorname{Im} \lambda_0 = a_0 T + B + \frac{C}{T} + \mathcal{O}(T^{-3}), \qquad a_0 = -fv, \quad B = -\pi, \quad C = -\frac{\pi c}{24\, v}, \tag{A.16}$$

This is the form of Eq. (6) fitted in Section 7. The extensive term $f\beta$ fixes a_0 and the factor -1 in the logarithm fixes B, while the coefficient C contains the central charge. Taking the difference between two logarithmic eigenvalues removes the common terms, including the Casimir contribution, and gives Eq. (7). For small β_0, the moduli vary only weakly with time, leading to the emergent dual unitarity tested in Section 7.

B Truncation schemes

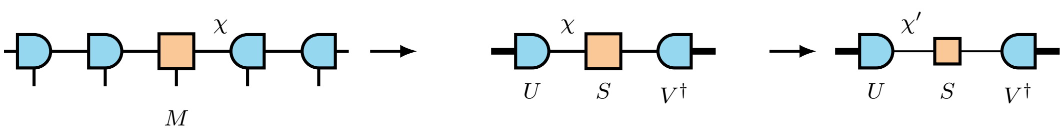
Truncation keeps the bond dimension bounded during transverse contraction. We now examine the two schemes introduced in Section 3.4, based on the RDM and the RTM. We first explain why Schmidt-value truncation is optimal for a single state. We then use gauge transformations to distinguish physical spectra from representation-dependent ones and identify the approximations involved in the RTM scheme.

Consider a cut that divides the chain into a block $\mathcal{A}$ and its complement $\mathcal{B}$. Grouping the physical indices on each side into a composite index represents the wavefunction as a matrix. Its Singular Value Decomposition (SVD) gives the Schmidt decomposition

$$|\psi\rangle = \sum_{\alpha=1}^{\chi} s_\alpha \, |\alpha\rangle_{\mathcal{A}} \otimes |\alpha\rangle_{\mathcal{B}}\,, \qquad s_\alpha \geq 0, \quad \sum_\alpha s_\alpha^2 = 1, \tag{B.1}$$

where the states on each side form orthonormal bases. In mixed canonical form, the Schmidt coefficients appear as a diagonal matrix on the cut bond. The corresponding RDM is diagonal in the same basis, with eigenvalues $p_\alpha = s_\alpha^2$ that determine the entanglement across the cut. Reducing the bond dimension from χ to $\chi' < \chi$, as shown in Figure 12, retains the χ' largest Schmidt coefficients. The Eckart–Young–Mirsky theorem establishes that this is the closest state to $|\psi\rangle$ among all states of Schmidt rank at most χ'. Its squared norm error is the discarded weight,

$$\varepsilon = \big\| \, |\psi\rangle - \big|\tilde{\psi}\big\rangle \big\|_2^2 = \sum_{\alpha > \chi'} s_\alpha^2. \tag{B.2}$$

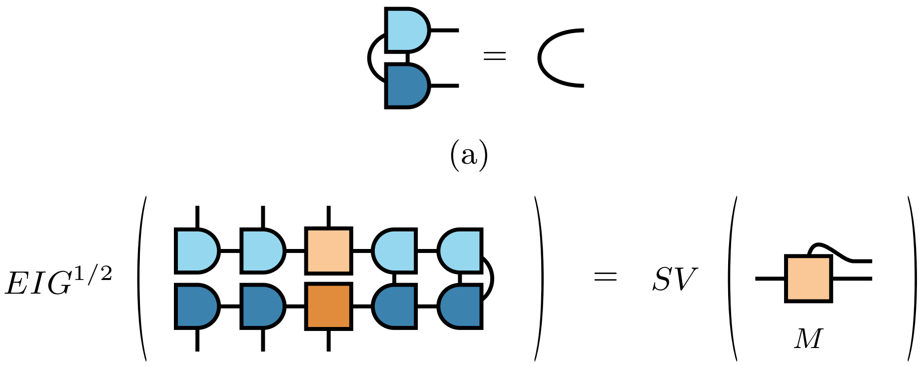


(a) canonical form (b) we group the legs to form matrices (c) we truncate

Figure 12: **Schmidt truncation at a bond.** (a) Chain in mixed canonical form, with the orthogonality centre M at the cut. (b) Grouping the indices on either side into composite indices (thick lines) gives the matrix decomposition $M = USV^\dagger$. (c) Truncated decomposition, with the internal dimension reduced from χ to χ'.

B.1 The RDM scheme

In canonical form, the RDM scheme obtains this truncation from a local tensor, as shown in Figure 13.



(a)

(b)

Figure 13: **Local RDM truncation in canonical form.** (a) Contracting a canonical tensor with its conjugate over the physical index and one virtual index gives the identity on the remaining index. Lighter and darker shades distinguish the tensor and its conjugate. (b) Reduced density matrix at a cut, with the left block open and the right block traced. The surrounding isometries contract to identities, so the eigenvalues of the full RDM are the squared singular values of the centre tensor M.

The truncation data can therefore be extracted from a local $\chi \times \chi$ environment matrix M_A rather than from the exponentially large global object. For a bipartite cut,

$$\rho_B = U_B \, M_A \, U_B^{\dagger}, \tag{B.3}$$

where U_B is unitary because it is constructed from left-orthogonal tensors. Equation (B.3) is a similarity transformation, so $\mathrm{eig}(\rho_B) = \mathrm{eig}(M_A)$. The local environment therefore has the same eigenvalues as the full RDM.

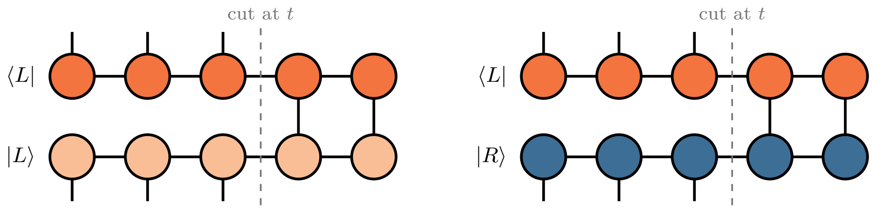
This construction relies on the RDM being Hermitian and positive. Its eigenvalues are the squared Schmidt coefficients and can therefore be ordered unambiguously, while the Eckart–Young–Mirsky theorem provides a variational error bound. Neither property extends directly to a non-Hermitian transition matrix.

B.2 The RTM scheme and its limits

In the transverse picture, the transfer matrix is non-Hermitian and its dominant left and right fixed points are distinct. The corresponding truncation object is the RTM in Eq. (11), constructed from both states. Its eigenvalues can be complex and do not have the Schmidt interpretation described above. Moreover, a gauge transformation can redistribute weight between components that contribute to $\langle L|R\rangle$ and those that do not [45]; the magnitude of a component in either state separately is therefore not a gauge-invariant measure of its contribution to the overlap. Figure 14 compares the RDM and RTM at the same temporal cut.

The tensor representation of either object is not unique. Let $A^{[n]}$ and $A^{[n+1]}$ be the MPS tensors on two sites joined by a virtual bond. We may insert $\mathbb{1} = XX^{-1}$ on this bond and absorb one factor into each tensor,

$$A^{[n]} \to A^{[n]}X, \qquad A^{[n+1]} \to X^{-1}A^{[n+1]}, \tag{B.4}$$



(a) $\rho_L = \mathrm{Tr}_{T-t}\,|L\rangle\,\langle L|$ (b) $\mathcal{T}_t = \mathrm{Tr}_{T-t}\,|R\rangle\,\langle L|\ /\ \langle L|R\rangle$

Figure 14: **Possible truncation objects at a temporal cut.** Sites after the cut are traced over, while the physical legs before the cut remain open and form the matrix indices. (a) The RDM pairs a temporal state with its conjugate and is Hermitian and positive. (b) The RTM pairs the distinct fixed points $|R\rangle$ and $\langle L|$; it is generally non-Hermitian and can have complex eigenvalues.

without changing the represented state, because the two factors cancel when the bond is contracted [6, 7]. The physical overlap is invariant under this transformation,

$$\langle L|R\rangle \;=\; \langle L|\,XX^{-1}\,|R\rangle \;=\; \left\langle \tilde{L}\middle|\tilde{R}\right\rangle, \tag{B.5}$$

but the spectra used for truncation need not be. Their transformation depends on whether the state across the cut is paired with its conjugate or with an independent left state.

The RDM pairs the state with its conjugate. Conjugating the first transformation in Eq. (B.4) gives $A^{[n]*} \to A^{[n]*}X^*$. Thus, where the ket carries X, the bra carries $X^\dagger$, and

$$\rho_L \;\to\; X^\dagger \rho_L X. \tag{B.6}$$

The factors do not cancel unless X is unitary, because $X^\dagger X \neq \mathbb{1}$ in general. Equation (B.6) is a congruence transformation and does not preserve the eigenvalues. A gauge transformation can therefore move components that contribute to the overlap into the tail of the isolated RDM spectrum, where they may be discarded by truncation.

The RTM instead pairs $|R\rangle$ with the independent state $\langle L|$. The left tensors absorb X^{-1} on the other side of the same bond, following the second transformation in Eq. (B.4). Consequently,

$$\mathcal{T}_t \;\to\; X^{-1}\mathcal{T}_t X. \tag{B.7}$$

The factors now cancel because $X^{-1}X = \mathbb{1}$. Equation (B.7) is a similarity transformation for any invertible X and therefore preserves the eigenvalues. Figure 15 shows the corresponding network contractions for both schemes.

The spectrum of a single temporal-state RDM therefore depends on the chosen representation, whereas the RTM spectrum is invariant for the pair $\langle L|\,, |R\rangle$. Fixing a canonical gauge makes the RDM truncation well defined, allowing it to be used as a complementary numerical scheme despite this representation dependence.

On the other hand, the local reduction of the RTM differs significantly from that of the RDM. For two distinct states, the construction leading to Eq. (B.3) instead gives

$$\tau_B = U_B\,M_A\,V_B, \qquad U_B \neq V_B^\dagger, \tag{B.8}$$

which is not a similarity transformation. Its eigenvalues are therefore not preserved: $\mathrm{eig}(\tau_B) \neq \mathrm{eig}(M_A)$. However, U_B and V_B remain isometries and preserve singular values,

$$\mathrm{sv}(\tau_B) = \mathrm{sv}(M_A). \tag{B.9}$$

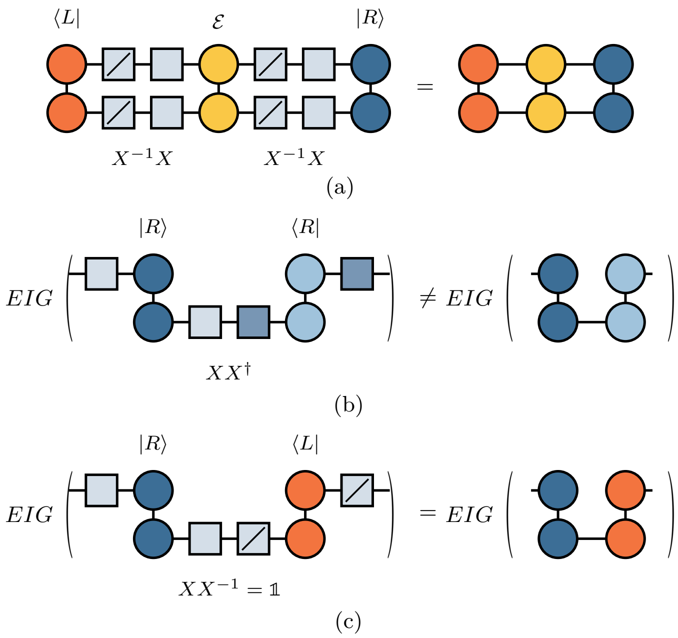


Figure 15: **Gauge transformation of the RDM and RTM.** (a) Inserting $X^{-1}X$ on a virtual bond changes the tensors but not the full contraction. In (b) and (c), the lower time site is contracted while the upper site remains open. (b) For the RDM, the gauge factors combine as $XX^\dagger \neq \mathbb{1}$ and produce the congruence $X\rho X^\dagger$, which changes the spectrum. (c) For the RTM, they combine as $XX^{-1} = \mathbb{1}$ and produce the similarity transformation $X\mathcal{T}_t X^{-1}$, which preserves the spectrum.

Thus, the local environment and the full RTM have the same singular values. These are the invariant quantities available from the local reduction and provide the truncation criterion we use in practice.

This local reduction requires a canonical form adapted to the left–right pair. As shown in Figure 16, the canonical condition is imposed on the pair $(\langle L|, |R\rangle)$ rather than on a state and its conjugate. Tensors away from the cut then contract to identities, leaving a local decomposition with the singular values of the full RTM. The two schemes consequently have the same local structure and comparable cost at fixed bond dimension.

The potential reduction in bond dimension arises because the RTM measures relevance through the left–right overlap. The method uses a biorthogonal decomposition of $|R\rangle\langle L|$ and retains the components with largest-magnitude eigenvalues [27]. For the transfer matrices considered here, this can reach a given accuracy with a smaller bond dimension than the RDM scheme. The required RTM rank is bounded above by the operator entanglement of the bipartition [27].

The local truncation criterion is expressed through singular values, whereas the physical overlap is determined by the trace of $\mathcal{T}_t$. These quantities are not equivalent. Singular values satisfy $\mathrm{sv}(M) = \sqrt{\mathrm{eig}(MM^\dagger)}$ and coincide with the eigenvalues for a Hermitian positive RDM, but not for a non-Hermitian RTM. The trace $\mathrm{Tr}\,\mathcal{T}_t = \sum_i \lambda_i$ is a signed sum and is not controlled directly by singular-value magnitudes. For example, eigenvalues $+1$ and -1 both have unit magnitude but cancel in the trace.

The RTM truncation therefore lacks the variational bound available for an RDM, and the same criterion need not be reliable for a generic non-Hermitian matrix. Cancellation would require retained and discarded components of comparable size, which the quickly decaying spectra of the model studied here at the critical point do not provide. The

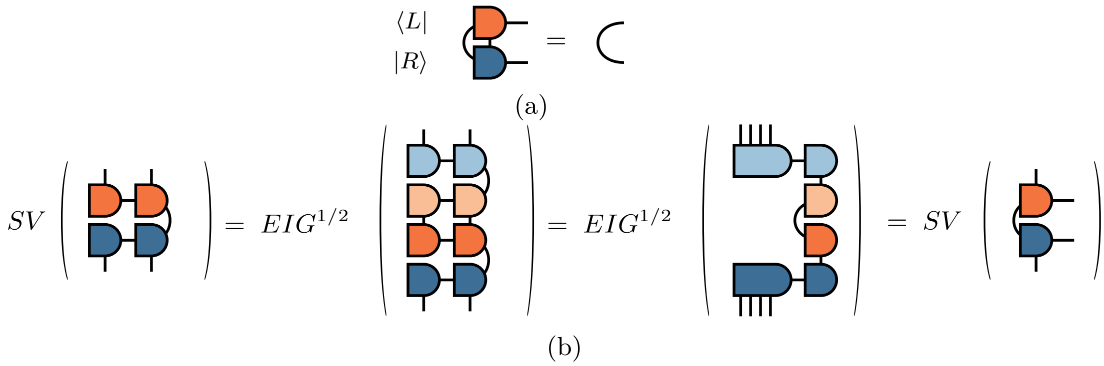


Figure 16: **Local RTM truncation in generalised canonical form.** (a) The contraction pairs a tensor of $\langle L|$ with the corresponding tensor of $|R\rangle$ rather than with its conjugate. (b) Imposing this condition along the chain reduces the environment to the tensors at the cut, whose singular values equal those of the RTM. The construction remains local but no longer has a variational error bound.

different guarantees are reflected in the convergence: the RDM scheme generally converges smoothly, whereas the RTM result can vary irregularly.

Both schemes make one further approximation. At each stage of the sweep, the truncation uses the current local environment without including the uncontracted remainder of the network. Translation invariance reduces the resulting error because every subsequent bulk column applies the same operator and can regenerate discarded components of the leading subspace.

C Why the NNN coupling breaks integrability

Mapping a one-dimensional spin chain to fermions provides a direct test of whether it reduces to free fermions. Under the JW transformation, a quadratic fermionic Hamiltonian can be reduced to independent normal modes and diagonalised exactly. Terms of fourth order instead couple these modes and remove this free-fermion integrability. We apply this criterion to Eq. (13).

The mapping must account for the different algebras of spins and fermions: Pauli operators on different sites commute, whereas fermionic operators anticommute. The JW transformation achieves this through a parity string over all sites to the left of the operator. In the convention of Eq. (13), with the transverse field along σ^x and the order parameter along σ^z, it is [46, 47]

$$\sigma_j^x = 1 - 2n_j, \qquad \sigma_j^z = (c_j + c_j^\dagger)\, K_j, \qquad K_j = \prod_{k<j} (1 - 2n_k), \tag{C.1}$$

where c_j annihilates a spinless fermion, $n_j = c_j^\dagger c_j$, and K_j measures the fermion parity to the left of site j.

Two identities determine how these strings cancel. Each parity factor squares to the identity, $(1-2n_k)^2 = 1$, and hence $K_j^2 = 1$. Moreover, $1 - 2n_k = (-1)^{n_k}$ is even in the fermionic operators and commutes with operators on other sites, while on the same site

$$(c + c^\dagger)(1 - 2n) = c^\dagger - c. \tag{C.2}$$

The transverse field maps directly to the chemical-potential term $\sigma_j^x = 1 - 2c_j^\dagger c_j$. For the NN interaction, using $K_{j+1} = K_j(1-2n_j)$ gives

$$\begin{aligned}\sigma^z_j\sigma^z_{j+1} &= (c_j + c_j^\dagger)K_j\,(c_{j+1}+c_{j+1}^\dagger)K_j(1-2n_j)\\ &= (c_j + c_j^\dagger)(c_{j+1}+c_{j+1}^\dagger)\,K_j^2\,(1-2n_j)\\ &= (c_j + c_j^\dagger)(1-2n_j)\,(c_{j+1}+c_{j+1}^\dagger)\\ &= (c_j^\dagger - c_j)(c_{j+1}+c_{j+1}^\dagger).\end{aligned} \tag{C.3}$$

The string commutes with the operators on site $j+1$ because it is even in the fermions. The remaining factor $(1-2n_j)$ is then combined with the operator on site j using Eq. (C.2). No parity factor survives, so the interaction is quadratic. Together with the field term, this makes the $p=0$ chain a free-fermion model that can be diagonalised exactly by a Bogoliubov transformation [35]. At criticality, its continuum limit is the free Majorana CFT with $c=\frac{1}{2}$ [23].

For the NNN interaction, the parity string crosses the intervening site. Since $K_{i+2} = K_i(1-2n_i)(1-2n_{i+1})$, the cancellation leaves two parity factors:

$$\begin{aligned}\sigma^z_i\sigma^z_{i+2} &= (c_i + c_i^\dagger)K_i\,(c_{i+2}+c_{i+2}^\dagger)K_i(1-2n_i)(1-2n_{i+1})\\ &= (c_i + c_i^\dagger)(1-2n_i)\,(1-2n_{i+1})\,(c_{i+2}+c_{i+2}^\dagger)\\ &= (c_i^\dagger - c_i)\,(1-2n_{i+1})\,(c_{i+2}+c_{i+2}^\dagger).\end{aligned} \tag{C.4}$$

The factor on site i is absorbed as in the NN case, but the factor on site $i+1$ remains because the bond contains no spin operator there. The hopping and pairing between sites i and $i+2$ therefore depend on the occupation of the intervening site.

Expanding the remaining parity factor separates the quadratic and quartic contributions:

$$\sigma^z_i\sigma^z_{i+2} = (c_i^\dagger - c_i)(c_{i+2}+c_{i+2}^\dagger) - 2\,(c_i^\dagger - c_i)\,c_{i+1}^\dagger c_{i+1}\,(c_{i+2}+c_{i+2}^\dagger). \tag{C.5}$$

The second term contains four fermionic operators. The self-dual interaction likewise contains a density–density contribution,

$$\sigma^x_j\sigma^x_{j+1} = 1 - 2n_j - 2n_{j+1} + 4\,c_j^\dagger c_j c_{j+1}^\dagger c_{j+1}. \tag{C.6}$$

Both terms proportional to p therefore contain four-fermion interactions. These interactions couple the normal modes of the $p=0$ chain, so Wick's theorem no longer reduces correlation functions to products of two-point functions [48], and no single-particle transformation diagonalises the Hamiltonian, since a linear transformation of the modes preserves the order of the interaction terms. Thus $p \neq 0$ removes the quadratic free-fermion structure responsible for integrability at $p=0$.

D Equilibrium measurements

The dynamical analysis in Section 7 requires two equilibrium quantities. The equilibrium central charge provides an independent reference for the dynamical estimate, while the sound velocity converts transfer-matrix phase gaps into scaling dimensions. We determine both quantities here.

D.1 Finite-size scaling of the mid-chain entropy

Section 4.1 extracts the central charge from the entanglement profile at a fixed system size. We obtain an independent estimate from the size dependence of the entropy across the mid-chain cut. At this cut, the chord variable is maximal, $W(L/2, L) = \ln(2L/\pi)$, and Eq. (14) becomes

$$S_{\mathrm{vN}}(L/2) = \frac{c}{6}\ln L + k'. \tag{D.1}$$

A linear fit against $\ln L$ therefore has slope $c/6$. For $N = 40$–200, the uncorrected fit gives $c = 0.511$ at $p = 0.1$ and $c = 0.514$ at $p = 0.5$. The same ground states give the Rényi-2 entropy used in the temporal calculation, whose logarithmic coefficient is $c/8$. The corresponding uncorrected estimates, $c = 0.584$ and 0.588, lie appreciably above the Ising value. Figure 17(a) shows both estimators.

This discrepancy is caused by finite-size corrections associated with the conical singularity of the replica geometry. Operators that are relevant at this singularity produce corrections proportional to $\ell^{-x/n}$ for an interval anchored at the boundary, where x is the scaling dimension of that operator [49]. The mid-chain cut of an open chain has this geometry, so we fit

$$S_n(L/2) = \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\ln L + k_n + A_n L^{-x/n}. \tag{D.2}$$

with A_n a non-universal amplitude. Which operator appears at the singularity is not known in advance, so we determine its dimension from the data: fits with x free return values close to 1 at both n, although a three-parameter fit to nine sizes does not determine it sharply. We therefore fix $x = 1$. The corrected estimates are $c = 0.500$ at both couplings for the von Neumann entropy and $c = 0.508$ and 0.505 for the Rényi-2 entropy, and the largest residual falls by two orders of magnitude in every case. Figure 17(b) shows the corrections that the fits absorb.

The factor x/n explains why the uncorrected estimates behave differently. For the same x, the correction to the von Neumann entropy decays as L^{-x}, whereas that of the Rényi-2 entropy decays as $L^{-x/2}$. The former is therefore already within three per cent of $\frac{1}{2}$, while the latter requires the correction term. The analogous temporal correction scales as $\Delta S_n \propto T^{-1/n}$ [19] and is included in Section 7.1.

The finite-size calculations use fifteen DMRG sweeps, a bond-dimension ramp up to 400, and a truncation cutoff of 10^{-10}, beginning with noise-assisted sweeps. The $N = 400$ chord fit in Section 4.1 instead uses fifty sweeps, a maximum bond dimension of 1000, and a cutoff of 10^{-12}.

D.2 The sound velocity from the equilibrium dispersion

We obtain the sound velocities used in Section 4.2 from the low-momentum dispersion, $\omega = v|k|$. On a periodic chain of N sites, the allowed momenta are $k_m = 2\pi m/N$, so a finite-size estimate follows from the lowest excitations at $m = 0$ and $m = 1$. These states must be identified within the same symmetry sector before their energy difference can be interpreted as a dispersion.

We therefore diagonalise the Hamiltonian in Eq. (13) at $\lambda = 1$ together with its translation symmetry. The one-site translation operator T has eigenvalues $e^{2\pi i m/N}$ and determines the momentum index m. It commutes with the Hamiltonian, so every level carries a definite momentum.

Degenerate eigenspaces require additional care because numerical diagonalisation returns arbitrary linear combinations of states with the same energy. We group levels whose energies differ by less than 10^{-8} and, within each cluster, diagonalise T, which restores the labels. A shift by N sites returns the ring to itself, so $T^N = \mathbb{1}$ and the eigenvalues of T are the Nth roots of unity, $e^{2\pi i m/N}$ with m an integer. An eigenstate therefore has

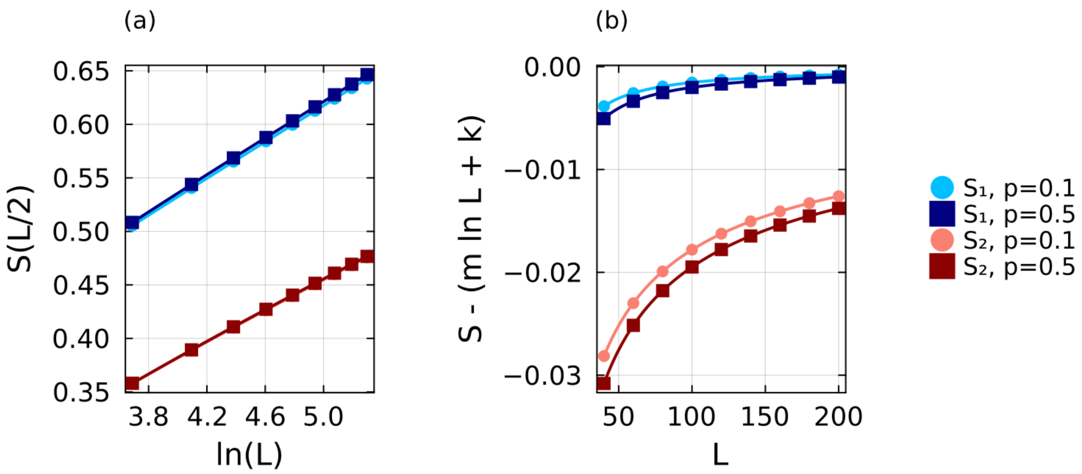


Figure 17: **Finite-size scaling of the mid-chain entropy.** (a) The von Neumann entropy S_1 and Rényi-2 entropy S_2 against $\ln L$ at $p = 0.1$ and $p = 0.5$, with the fits of Eq. (D.2). The corrected estimates are $c = 0.500$ at both couplings from S_1, and $c = 0.508$ and 0.505 from S_2. The correction changes the curves by less than 10^{-3} and is not visible on this scale. (b) The same data after subtracting the fitted logarithmic contribution, which isolates the finite-size correction.

$\langle T\rangle = e^{2\pi i m/N}$, and its phase gives m directly. The computed phase is not exactly a multiple of $2\pi/N$, so we round it to the integer it must be, and assign

$$m = \text{round}\left[\frac{N}{2\pi}\arg\langle T\rangle\right]. \tag{D.3}$$

With arg taken in $(-\pi, \pi]$ the index is signed, and the momentum is $k_m = 2\pi m/N$. Only the ten lowest levels are computed.

The finite-size velocity is then

$$v(N) = \frac{E_1(2\pi/N) - E_1(0)}{2\pi/N} = \frac{N}{2\pi}\big[E_1(2\pi/N) - E_1(0)\big]. \tag{D.4}$$

Nothing above uses the parity $P = \prod_i \sigma_i^x$, although it commutes with the Hamiltonian too. Ignoring it is safe only if the lowest states at $m = 0$ and $m = 1$ happen to share a parity, since otherwise their energy difference mixes two branches. We checked it directly. Labelling both symmetries at once means diagonalising $c_1 T + c_2 P$ inside each degenerate cluster, so that neither label spoils the other; recomputing from the odd-parity states alone then leaves the velocity unchanged to every digit we store, at $N = 10$ and $N = 12$ and for every coupling up to $p = 0.5$. The two states already share a parity, so we do not project. This is the discrete form of the low-momentum slope $\mathrm{d}\omega/\mathrm{d}k$.

We validate the conversion from a spectral gap to a scaling dimension using the anisotropic XY chain,

$$H_{\text{XY}} = -\sum_j \left[\frac{1+\gamma}{2}\,\sigma_j^x\sigma_{j+1}^x + \frac{1-\gamma}{2}\,\sigma_j^y\sigma_{j+1}^y\right] - \lambda\sum_j \sigma_j^z, \tag{D.5}$$

written with the coupling on σ^x and the field on σ^z, the opposite convention to Eq. (13); at $\gamma = 1$ it reduces to the transverse-field Ising chain. The transformation of Appendix C maps it to free fermions with dispersion

$$\varepsilon(k) = 2\sqrt{(\lambda - \cos k)^2 + \gamma^2\sin^2 k}. \tag{D.6}$$

At $\lambda = 1$ the gap closes at $k = 0$, which is what puts the chain at criticality. Expanding there with $1 - \cos k \simeq k^2/2$ and $\sin k \simeq k$ gives $\varepsilon \simeq 2|k|\sqrt{\gamma^2 + k^2/4}$, so $\varepsilon \to 2\gamma|k|$ as $k \to 0$ and the velocity is exactly $v(\gamma) = 2\gamma$. At $\gamma = 1$ this returns the Ising value $v = 2$ used throughout. The leading boundary dimension remains $x_1 = \frac{1}{2}$. The test runs through Eq. (7), which gives the first gap as $\mathrm{Im}(\lambda_1 - \lambda_0) = \pi x_1/(vT)$. Its leading behaviour is thus $1/T$; the terms linear in T and constant in T are common to both eigenvalues and cancel in the difference, so the next correction is $1/T^3$. We therefore fit

$$\mathrm{Im}(\lambda_1 - \lambda_0) = \frac{a_1}{T} + \frac{a_3}{T^3}, \tag{D.7}$$

whose leading coefficient is $a_1 = \pi x_1/v$. Both quantities on the right are known: the initial product state realises the same free boundary as in Section 2.1, so $x_1 = \frac{1}{2}$, and $v = 2\gamma$ from the expansion of Eq. (D.6), which gives the exact value $a_1 = \pi/(4\gamma)$. Read the other way round, $x_1 = v\, a_1/\pi$ turns a measured slope into a dimension. As shown in Table 2, using the correct velocity gives $x_1 = 0.461$, 0.497, and 0.496. Including the coupling-dependent velocity therefore keeps the extracted dimension close to $\frac{1}{2}$.

γ	$v = 2\gamma$	a_1	$\pi/4\gamma$	x_1 from $v = 2\gamma$
0.5	1	1.449	1.571	0.461
1.0	2	0.781	0.785	0.497
1.5	3	0.520	0.524	0.496

Table 2: **Validation on the anisotropic XY chain.** The first spectral gap is fitted as $\mathrm{Im}(\lambda_1 - \lambda_0) = a_1/T + a_3/T^3$ over the converged times: $T = 2, 3, 4$ for $\gamma = 0.5$ and 1.0, and $T = 2, 4$ for $\gamma = 1.5$. The exact coefficient is $a_1 = \pi/(4\gamma)$ for $x_1 = \frac{1}{2}$. The final column converts the fitted coefficient using the correct velocity $v = 2\gamma$.

Two independent checks validate the finite-ring calculation. The Hamiltonian used here is constructed directly through bit operations, independently of the MPO, and their full spectra at $N = 8$ agree to within 7×10^{-14}. The second check turns our velocity into a central charge. For a periodic chain the ground-state energy per site is $f(L) = f_\infty - \pi c\, v/(6L^2)$ [43], and a finite difference between L and $L-1$ removes the unknown f_∞,

$$c = -\frac{6\,[\,f(L) - f(L-1)\,]}{\pi v\,\left(L^{-2} - (L-1)^{-2}\right)}. \tag{D.8}$$

Evaluated on our own ground-state energies and velocities at $N = 16$, this gives $c = 0.504$ at $p = 0$ and $c = 0.525$ at $p = 0.5$, consistent with previously reported values for this model [20].

The accessible ring sizes are far from the thermodynamic limit: at $p = 0$ the $N = 18$ estimate is 1.9899 against the exact value 2, and reaching three-digit accuracy directly would need about $N = 57$. We therefore extrapolate, and fix the form of the correction at the integrable point, where the answer is known. Writing the error as $2 - v(N) \propto N^{-\alpha}$, the rescaled quantity $[2 - v(N)]\, N^\alpha$ is constant to within half a per cent for $\alpha = 2$ alone, drifting downwards for $\alpha = 1$ and upwards for $\alpha = 3$. Extrapolating with those two powers misses the exact velocity by 2×10^{-2} and 6×10^{-3}, against 6×10^{-5} for $1/N^2$. A further D/N^4 term is not constrained by five sizes and does not improve the extrapolation.

We therefore fit

$$v(N) = v_\infty + \frac{C}{N^2} \tag{D.9}$$

at every coupling and use v_∞ throughout; Table 3 lists the data and fit parameters, and Figure 18 shows the extrapolations. What this buys is clearest where it can be checked:

at $p = 0$ the extrapolated value lies within 6×10^{-5} of the exact result, against 1.0×10^{-2} for the raw $N = 18$ estimate. It also matters more as the coupling grows, since $|C|$ rises from 3.27 at $p = 0$ to 140.7 at $p = 1.5$.

p	$N = 10$	$N = 12$	$N = 14$	$N = 16$	$N = 18$	v_∞	C
0.0	1.9673	1.9772	1.9833	1.9872	1.9899	1.99994	−3.27
0.1	2.5707	2.6007	2.6191	2.6312	2.6396	2.67016	−9.97
0.2	3.1565	3.2073	3.2387	3.2594	3.2738	3.32561	−16.96
0.3	3.7247	3.7972	3.8423	3.8722	3.8931	3.96723	−24.34
0.4	4.2756	4.3707	4.4305	4.4703	4.4981	4.59585	−32.16
0.5	4.8096	4.9284	5.0038	5.0543	5.0897	5.21214	−40.46
1.0	7.2534	7.5020	7.6674	7.7824	7.8653	8.12650	−88.23
1.5	9.4140	9.7946	10.0594	10.2504	10.3921	10.79924	−140.69

Table 3: **Finite-size velocities and thermodynamic extrapolations.** Values of $v(p)$ at $\lambda = 1$ for periodic chains of $N = 10$–18, with the parameters of the fit $v(N) = v_\infty + C/N^2$. The largest residual increases from 1.1×10^{-5} at $p = 0$ to 2.8×10^{-2} at $p = 1.5$. The conformal analysis uses the extrapolated values v_∞.

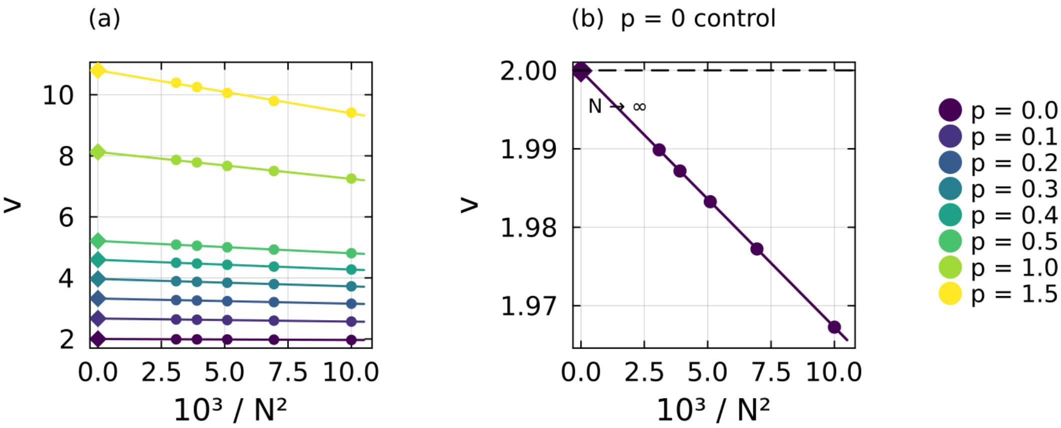


Figure 18: **Extrapolation of the sound velocity.** (a) Finite-size estimates of v against $1/N^2$ at each coupling, with fitted lines and thermodynamic intercepts (diamonds). (b) The integrable point on an expanded scale. The estimates are linear in $1/N^2$, and the intercept $v_\infty = 1.99994$ agrees with the exact value $v = 2$ (dashed line).

Over the range studied the extrapolated velocity is close to linear in p, $v_\infty \simeq 2.02 + 6.42p$, which is the trend shown in Figure 5. The linearity is only approximate: the slope between successive couplings falls steadily from 6.70 near $p = 0$ to 5.35 between $p = 1.0$ and $p = 1.5$, so the growth flattens at stronger coupling, and a fit at fixed system size still drifts with N. We therefore treat the relation as a description of the trend, and every conversion in this thesis uses the separately extrapolated $v_\infty(p)$.

E Construction of the time-evolution MPO

This appendix describes the time-evolution MPO used in Section 5. We give its construction from the Hamiltonian MPO, the explicit local tensor of the second-order scheme, and numerical comparisons with the lower-order alternatives.

The construction begins from the Hamiltonian represented as an MPO. We supply the Hamiltonian as a symbolic sum of operator strings, from which the library assembles the FSM of Section 5.1. At each bond, it collects the terms that cross the bond, arranges their coefficients into a matrix, and compresses this matrix with an SVD. The retained components form the minimal set of memory channels required to carry the interactions across the bond. We denote their number by χ_H to distinguish it from the bond dimension χ used to truncate the temporal states. For the ANNNI-type Hamiltonian, the compression gives $\chi_H = 3$, corresponding to the three intermediate states in Eq. (15), and hence $D_W = \chi_H + 2 = 5$. It also determines the four blocks in that equation: D contains the on-site term, C initiates an interaction, B terminates it, and A propagates it across a site.

We construct the local evolution tensor with `ITensorExpMPO.jl` [50], which is built on `ITensors.jl` [51] and implements the operators W^{I} and W^{II} of Ref. [41]. Under the single-layer application considered here, neither operator is second-order accurate, as shown below. We therefore added the second-order cluster construction of Ref. [40]. The implementation is available in a fork of the package [52], together with its documentation.

E.1 The second-order *VD2* construction

We seek a local tensor $W(\tau)$, with $\tau = -i\delta t$, whose product along the chain approximates $e^{\tau H}$. Writing the Hamiltonian as $H = \sum_x H_x$, the expansion is

$$e^{\tau H} = \mathbb{1} + \tau \sum_x H_x + \frac{\tau^2}{2} \sum_{x,y} H_x H_y + \dots \tag{E.1}$$

Products of terms with disjoint support commute and factorise. A repeated local tensor therefore generates them without additional memory channels. The non-trivial contributions are products of overlapping terms. Since the corresponding blocks act on a common site, they need not commute, and both $H_x H_y$ and $H_y H_x$ must appear with the weights prescribed by Eq. (E.1). The accuracy of an evolution MPO is determined by which of these overlapping products it retains.

The constructions W^{I} and W^{II} encode the products through the virtual index [41]. Each memory channel is represented by an auxiliary hard-core boson whose occupation records whether a Hamiltonian term crosses the bond. Restricting the bond to at most one boson retains products in which no two terms cross the same bond. The more restrictive operator W^{I} keeps only disjoint-support products. The operator W^{II} also retains products that overlap on one site and consequently reproduces the on-site field to all orders. Both have virtual bond dimension $1 + \chi_H$, which is four for the present Hamiltonian.

Neither operator retains products in which two terms cross the same bond. This omission enters at order δt^2, making a single application first-order accurate for both the NNN model and the strictly NN Ising chain, as verified in Appendix E.2. Their errors nevertheless differ substantially. The operator W^{I} also omits products involving the on-site field, which occur throughout the chain, whereas W^{II} retains them. At $\delta t = 0.1$, this reduces the error by approximately a factor of five.

Second-order accuracy can be restored either by composing several stages [41] or by retaining the missing overlapping pairs in a single application. A staged composition is efficient in conventional time evolution because the sub-steps together advance one interval δt. In the transverse contraction, however, the stages must be blocked into a single repeated column. Two stages give a local physical dimension $4^2 = 16$ after rotation, compared with 13 for the second-order construction used here.

The *VD2* construction organises the expansion into connected clusters of overlapping terms [40]. Retaining clusters of up to n terms gives a one-step error of order δt^{n+1}. The first-order member is W^{II}, while the second-order member used here reproduces every connected pair. Representing two interactions that cross the same bond requires a doubled-memory sector of dimension χ_H^2, in addition to the scalar and single-memory sectors. The resulting virtual bond dimension is

$$1 + \chi_H + \chi_H^2. \tag{E.2}$$

For the ANNNI-type model, this gives 13, compared with 4 for W^{I} and W^{II}. At $p = 0$, only the NN interaction remains and $\chi_H = 1$, giving bond dimensions 3 and 2, respectively.

Before combining the blocks, we distribute the time step asymmetrically between the interaction endpoints. The initiating block carries $t_C = \sqrt{\delta t}$ and the terminating block carries $t_B = \tau/t_C$, so a completed interaction acquires one factor of τ:

$$\tilde{D} = \tau D, \qquad \tilde{C} = t_C\, C, \qquad \tilde{B} = t_B\, B, \qquad \tilde{A} = A. \tag{E.3}$$

This split is a rescaling of the virtual index. The two factors cancel when an interaction begins and ends within the chain, so physical matrix elements are unchanged. The rescaling instead balances the tensor entries at small δt, avoiding a representation in which one endpoint carries the complete time-step factor.

The local representation of $e^{\tau H}$ is written in terms of symmetrised products of the scaled blocks. We denote by $\{X_1 \dots X_m\}$ the sum over their distinct orderings, with $\{XY\} = X \otimes Y + Y \otimes X$, $\{XX\} = X \otimes X$, and $\{XXY\} = XXY + XYX + YXX$. Here $\otimes$ denotes the Kronecker product on the virtual memory index and ordinary operator multiplication on the physical index. The numerical prefactors below are the factorial weights from the exponential series. The expressions follow the second-order cluster operator of Ref. [40], specialised to the blocks of Eq. (15).

The local tensor has a 3×3 block structure. The indices 1, 2, and 3 label the scalar, single-memory, and doubled-memory sectors, respectively:

$$W^{VD2} = \begin{pmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \\ W_{31} & W_{32} & W_{33} \end{pmatrix}, \tag{E.4}$$

with

$$W_{11} = \mathbb{1} + \tilde{D} + \tfrac{1}{2}\{\tilde{D}\tilde{D}\} + \tfrac{1}{6}\{\tilde{D}\tilde{D}\tilde{D}\}, \tag{E.5}$$

$$W_{12} = \tilde{C} + \tfrac{1}{2}\{\tilde{C}\tilde{D}\} + \tfrac{1}{6}\{\tilde{C}\tilde{D}\tilde{D}\}, \qquad W_{13} = \{\tilde{C}\tilde{C}\} + \tfrac{1}{3}\{\tilde{C}\tilde{C}\tilde{D}\}, \tag{E.6}$$

$$W_{21} = \tilde{B} + \tfrac{1}{2}\{\tilde{B}\tilde{D}\} + \tfrac{1}{6}\{\tilde{B}\tilde{D}\tilde{D}\}, \qquad W_{31} = \tfrac{1}{2}\{\tilde{B}\tilde{B}\} + \tfrac{1}{6}\{\tilde{B}\tilde{B}\tilde{D}\}, \tag{E.7}$$

$$W_{22} = \tilde{A} + \tfrac{1}{2}(\{\tilde{B}\tilde{C}\} + \{\tilde{A}\tilde{D}\}) + \tfrac{1}{6}(\{\tilde{C}\tilde{B}\tilde{D}\} + \{\tilde{A}\tilde{D}\tilde{D}\}), \tag{E.8}$$

$$W_{23} = \{\tilde{A}\tilde{C}\} + \tfrac{1}{3}(\{\tilde{A}\tilde{C}\tilde{D}\} + \{\tilde{C}\tilde{C}\tilde{B}\}), \qquad W_{32} = \tfrac{1}{2}\{\tilde{A}\tilde{B}\} + \tfrac{1}{6}(\{\tilde{A}\tilde{B}\tilde{D}\} + \{\tilde{B}\tilde{B}\tilde{C}\}), \tag{E.9}$$

$$W_{33} = \{\tilde{A}\tilde{A}\} + \tfrac{1}{3}(\{\tilde{A}\tilde{B}\tilde{C}\} + \{\tilde{A}\tilde{A}\tilde{D}\}). \tag{E.10}$$

The block W_{11} contains the on-site evolution through third order in the Taylor expansion. The blocks W_{12} and W_{13} open the single- and doubled-memory sectors, while W_{21} and W_{31} close them. The remaining blocks propagate interactions across the bond and combine $\tilde{A}$ with the required symmetrised products. The doubled-memory sector permits two interactions to cross the same bond simultaneously and contains nine of the thirteen virtual states. It therefore represents the overlapping pairs generated by the NNN coupling that are absent from W^{II}.

E.2 Comparing the two constructions in practice

We first examine how well each construction preserves the norm under repeated application. We apply each operator to $|X+\rangle^{\otimes N}$ on a chain of $N = 100$ sites at $p = 0.1$, using $\delta t = 0.05$ and truncating the state at 10^{-12} after every layer:

	W^{I}	W^{II}	*VD2*
virtual bond	4	4	13
$\|\psi\|$ after 1 layer	1.3514	1.1341	0.99995
$\|\psi\|$ after 8 layers	12.6146	3.0181	0.99923
$\|\psi\|$ after 16 layers	307.3900	10.8328	0.99751

The norm error accumulates rapidly under the two first-order constructions. After sixteen layers, W^{I} increases the norm by more than two orders of magnitude and W^{II} by approximately one order of magnitude. By contrast, the *VD2* norm remains within 2.5×10^{-3} of unity. A calculation at $T = 10$ contains approximately one hundred temporal layers, so the first-order operators are unsuitable at this step size.

Norm drift compares the accumulated stability but does not determine the largest usable time step at a fixed accuracy. We therefore compare the state error at a fixed physical time. For $N = 12$, the Hamiltonian can be diagonalised exactly. We evolve $|X+\rangle^{\otimes N}$ to $T = 1$ using $T/\delta t$ layers and compare the result with the exact state at $p = 0.1$:

	W^{I}		W^{II}		*VD2*	
δt	error	slope	error	slope	error	slope
0.2	2.01×10^{1}	–	2.54	–	1.09×10^{-1}	–
0.1	4.43	2.18	9.29×10^{-1}	1.45	2.44×10^{-2}	2.16
0.05	1.36	1.71	3.93×10^{-1}	1.24	5.93×10^{-3}	2.04
0.025	5.31×10^{-1}	1.35	1.80×10^{-1}	1.12	1.47×10^{-3}	2.01
0.0125	2.36×10^{-1}	1.17	8.65×10^{-2}	1.06	3.68×10^{-4}	2.00
0.00625	1.12×10^{-1}	1.08	4.24×10^{-2}	1.03	9.24×10^{-5}	2.00

The slope is the effective convergence order obtained from each error and the result at the preceding step size. Because the error contains higher-order terms, this estimate approaches the asymptotic order as δt decreases. The *VD2* slope is 2.04 at $\delta t = 0.05$ and remains 2.00 at smaller steps. The corresponding slopes for W^{I} and W^{II} are 1.71 and 1.24 at $\delta t = 0.05$, approaching 1.08 and 1.03 only at the smallest step. These results confirm second-order convergence for *VD2* and first-order convergence for the other two constructions.

At $p = 0.5$, W^{II} approaches the asymptotic regime more slowly, with its measured slope decreasing from 1.85 to 1.07 over the sampled steps. The stronger interaction therefore requires a smaller δt before the first-order term dominates. The *VD2* slope remains close to 2.00 throughout. At the next halved step, its error is 2.7×10^{-5} and the measured slope decreases to 1.77 because state truncation and the finite accuracy of the exact reference become comparable to the evolution error.

The larger virtual bond of *VD2* is compensated by the larger time step allowed at fixed accuracy. A first-order construction requires a smaller δt, which lengthens the temporal chain in inverse proportion and introduces an additional truncation at every new site. At $T = 10$, the temporal chain contains 104 sites for $\delta t = 0.1$ and 204 sites for $\delta t = 0.05$. The production calculations therefore use *VD2* with $\delta t = 0.1$.

F Implementation details of the block power method

Section 6.1 introduces the block power method: we propagate k left and k right temporal states through the transfer matrix, project the eigenproblem onto the subspace they span, and read the leading eigenvalues from the resulting $k \times k$ problem. This appendix describes what the implementation actually does at each of those steps, which numerical choices it exposes, and what we measured when we varied them. The routine is `block_transfer_eigs`, built on `ITransverse.jl` [28]; the code, the analysis notebooks and the cluster drivers are in the thesis repository [53].

F.1 The iteration as implemented

One iteration takes the current blocks $\{|R_j\rangle\}$ and $\{\langle L_j|\}$ and returns improved ones. It has four steps: apply, project, solve, and separate.

We first apply the transfer matrix to every member of the right block, and its transpose to every member of the left block, which is $2k$ operator–state products. Each product raises the temporal bond dimension, so every result is truncated immediately, back to the bond cap in force at that iteration. We then form the two $k \times k$ matrices of Eq. (19): the overlap matrix $S_{ij} = \langle L_i|R_j\rangle$ and the projected transfer matrix $M_{ij} = \langle L_i|\,\mathcal{E}\,|R_j\rangle$. Both are built from the same bilinear contraction, without complex conjugation, because the left states are already bras.

Solving $Mv = \theta Sv$ is where care is needed. As the iteration converges, every member of the block turns towards the dominant eigenvector, so the states become nearly parallel and S becomes nearly singular. Inverting it would divide by its smallest singular values, which at that point are numerical noise. We therefore use the Moore–Penrose pseudoinverse S^+, which inverts S only on the directions whose singular values exceed 10^{-12} times the largest one and maps the rest to zero. The near-dependent directions are discarded rather than amplified, and the generalised problem becomes an ordinary one,

$$W = S^+ M, \qquad W\, v_j = \theta_j\, v_j. \tag{F.1}$$

Diagonalising W gives the Ritz values θ_j, which approximate $\mu_0, \ldots, \mu_{k-1}$ and which we sort by decreasing modulus, together with the right coefficient vectors v_j. Collecting the v_j as the columns of V and the θ_j on the diagonal of Θ writes that diagonalisation as $W = V\Theta V^{-1}$.

The left block needs its own coefficients, and each one must belong to the same eigenvalue as the right coefficient it is paired with. Diagonalising the transposed problem separately does not guarantee this: the two diagonalisations return their eigenvalues in whatever order the algorithm produces, and inside a cluster of nearly equal eigenvalues there is no reliable way to decide which left vector belongs to which right vector. Instead we obtain the left coefficients from the decomposition we already have. We look for u with $u^\mathsf{T} M = \theta\, u^\mathsf{T} S$, and set $y^\mathsf{T} = u^\mathsf{T} S$. Using $S^+ S = \mathbb{1}$ on the retained directions, $u^\mathsf{T} M = y^\mathsf{T} S^+ M = y^\mathsf{T} W$, so the condition becomes $y^\mathsf{T} W = \theta\, y^\mathsf{T}$: the vector y is a left eigenvector of the same matrix W. Left eigenvectors of W are the rows of V^{-1}, since $V^{-1} W = \Theta V^{-1}$. Taking the j-th row and undoing the substitution gives

$$u_j = \left(S^+\right)^\mathsf{T} \left(V^{-1}\right)^\mathsf{T} e_j, \tag{F.2}$$

where e_j is the j-th unit vector. This u_j carries the same index j as v_j, so the pairing is fixed by construction rather than by matching values afterwards. The two sets are also biorthogonal in the metric S: $u_i^\mathsf{T} S v_j$ is the i-th row of V^{-1} times the j-th column of V, which is δ_{ij}.

The last step turns the coefficients back into states, which is the separation step of Algorithm 1: each new right state is the combination $\sum_j (v_i)_j \, |AR_j\rangle$ of the propagated right states, and likewise on the left. We form these sums by direct summation of the tensors, which is exact, and compress afterwards. Adding MPSs through their density matrices would be cheaper, but it becomes unstable precisely when two states are nearly parallel, which is the situation the block is in near a degeneracy. Each new state is then renormalised; if its norm has underflowed or overflowed, it is replaced by a fresh random state, so a lost direction is refilled rather than propagated as noise.

F.2 The eigenvalue-only mode

The separation step of Section F.1 is the fragile part of the iteration. It asks each block member to become one particular eigenvector, and that assignment is controlled by how well separated the eigenvalues are: when two of them approach, the coefficient matrix V that performs the separation becomes ill conditioned, and the states it produces are dominated by whatever noise the truncation left behind. The eigenvalues themselves do not share this weakness, because each one is obtained from a projection of the operator onto the subspace rather than from an individual vector.

When only the spectrum is needed, we therefore stop separating the states. After sorting the Ritz values by modulus we replace the coefficient matrices by orthonormal ones,

$$V \to Q_V, \qquad U \to Q_U, \qquad Q^\dagger Q = \mathbb{1}, \tag{F.3}$$

taking Q_V and Q_U from the QR decompositions of V and U. Because QR builds each column from the columns to its left, the first j columns of Q_V span the same subspace as the first j columns of V: the ordering by modulus is preserved, and the block still tracks the leading invariant subspace, but no member is asked to be a single eigenvector any more.

This substitution does not change the Ritz values. Replacing the blocks by any invertible recombination of their members, $|R_j\rangle \to \sum_i B_{ij} \, |R_i\rangle$ and $\langle L_j| \to \sum_i C_{ij} \, \langle L_i|$, sends the projected matrices of Eq. (19) to $S \to C^\mathsf{T} S B$ and $M \to C^\mathsf{T} M B$. The generalised problem $Mv = \theta S v$ then becomes $C^\mathsf{T} M B \, \tilde{v} = \theta \, C^\mathsf{T} S B \, \tilde{v}$ with $\tilde{v} = B^{-1} v$, which has the same θ. The QR rotation is one such recombination, so it moves the basis inside the subspace without moving the spectrum. What it does change is the conditioning of everything that follows: Q_V and Q_U are unitary, whereas V is not, so the states entering the next truncation are no longer amplified by a factor that grows as the eigenvalues approach.

Since the two modes differ only in that basis, and the basis does not affect the Ritz values, any difference between them has to come from the truncation applied to the states afterwards. Repeating the same ladder in both modes at $p = 0.1$, from the same random states and with every other setting held fixed, the leading modulus agrees to between 6×10^{-7} and 2×10^{-5} in relative terms over $T = 2$–8, and the ratio of the first subleading modulus to it agrees to between 10^{-8} and 10^{-4}; the members further down the block agree only to between 10^{-4} and 10^{-2}. Comparing the two production arms at $p = 0.1$ instead, which differ in more than the basis, gives the same ranges. The leading eigenvalue, which carries the central charge, is therefore insensitive to the choice, while the weakly converged members of the block are not.

Three safeguards run in both modes. Each state is renormalised after the separation step, and one whose norm has underflowed or overflowed is replaced by a fresh random state. The condition number of S is monitored: when it exceeds 10^{10}, two directions of the block have become numerically dependent, and the last member is replaced by a random state made biorthogonal to the others in the bilinear form of Eq. (19), which restores the rank without disturbing the converged members. Convergence is judged on the two leading Ritz values, as stated in Section 6.1. Each value is matched to its nearest predecessor before the change is measured, rather than compared at the same position in the sorted list, because three properties of this spectrum make the position unreliable. The moduli of the leading eigenvalues approach a common value as the evolution time grows, which is the emergent dual unitarity of Eq. (8); at $p = 0$ and $T = 20$ the three largest agree to three parts in a thousand, so ordering by modulus barely separates them. The transfer matrix is not Hermitian, so its eigenvalues are complex, and dual unitarity leaves their phases distinct: two eigenvalues of nearly equal modulus can lie far apart in the complex plane. And the Ritz values are recomputed at every iteration from states that have just been truncated, so a modulus difference this small is resolved only at the level of the truncation noise and the ordering can change from one iteration to the next. Comparing position by position would then read an exchange of two eigenvalues as a large change in one of them, and the iteration would never recognise that it had converged. The iteration stops when the matched change falls below 10^{-6}, or after 150 iterations in which it has not fallen further.

F.3 Validation against exact diagonalisation

At short evolution times, the rotated transfer operator has dimension below a few thousand and can be contracted into a dense matrix for exact diagonalisation. This comparison tests the operator construction, rotation, block iteration, and projected pencil against a result without TN truncation. Table 4 shows the results for $k = 4$; the evolution times are $T = 0.3$–1.5, before any near-degeneracy has developed, so the comparison validates the solver and the operator construction rather than the late-time regime. The two leading Ritz values agree with exact diagonalisation to between 10^{-7} and 10^{-13}.

operator	dimension	exact $\lvert\mu_0\rvert$, $\lvert\mu_1\rvert$	error
ANNNI-type, *VD2*, $p = 0.1$	343	0.9598, 0.3263	1.4×10^{-8}
ANNNI-type, W^{II}, $p = 0.1$	27	0.9758, 0.3292	4.6×10^{-13}
ANNNI-type, *VD2* bulk column, $p = 0.1$	2197	0.9405, 0.9082	3.6×10^{-7}
critical Ising	256	1.5488, 0.0611	4.2×10^{-10}

Table 4: **Validation against exact diagonalisation.** Comparison between the block method and dense diagonalisation of the same rotated transfer operator at short evolution times. The reported error is the largest absolute deviation in the moduli of the two leading Ritz values.

[not very clear what we are comparing, how, etc... right now reads very unclear and confusing because we are not giving simple explanations with the recipe we should always follow: explain why me need it, what it is, and how we implement it; here it does not have anything from it and it looks like we are seeing magic numbers and assuming the reader knows the context of the code (they do not have it and know nothing about the project, so we need to clarify and explain everything in great detail from the begining, without assuming the reader knows what we are doing, and with simple and clear explanations). this argument and comment applies to the rest of the subsections we are reworking]

F.4 Computational cost

Each block-power iteration contains application, projection, and de-mixing steps. The application step performs $2k$ tMPO–tMPS products, increasing the temporal bond dimension from χ to $D\chi$ before compression. The projection step evaluates the k^2 bilinear overlaps defining S and M; it does not increase the state dimensions, and the remaining $k \times k$ dense algebra is negligible[compare to what? be clear always on all the statements we are making, with clear explanations and clarifications, or with numerical evidence, otherwise it sounds like an unjustified argument with not rigours support, so no convincing]. The de-mixing step forms $2k$ linear combinations of k states. Exact direct-sum addition temporarily increases the bond dimension to $k\chi$ before compression, so this step determines both the peak memory and most of the runtime. For a representative production calculation with $\chi = 64$, $k = 4$, and $p = 0.1$, de-mixing and compression account for approximately 85% of each iteration[where does the percentage come from? it looks like a magic number if you just say it without justifying how you got that number; here it is better to say that given the sizes we are dealing with, we give a qualitative argument on this being the step which takes most of the memoroy/cpu power for example].

We reduce this cost in two ways. First, partial direct sums are compressed progressively back to the production bond dimension, while the final addition remains exact. Second, we shorten how long the iteration may run without improving, because late-time calculations can remain close to the convergence threshold for many iterations. Together, these changes reduce the runtime of the longest time points by approximately a factor of two and change $|\mu_0|$ by only 4×10^{-6} relative.[the two ways are explained in a really vague way, not very clear whatsoever, and assumes the reader know how the code works: explain everything from simple terms without assuming anything so that the reader understands everything in a self contained way]

F.5 Production options

A model-independent builder constructs the one-step propagator as an MPO (Appendix E), performs the 90° rotation from the spatial virtual bond to the temporal physical index, and returns the transfer-matrix MPO together with a seed tMPS whose physical dimension is inferred from the operator, so the same implementation treats W^{I}, W^{II}, *VD2*, and different models. Before iteration, the seed tensors are replaced by random complex tensors because a symmetry-restricted seed can converge to a subdominant sector. Four options modify the numerical procedure without changing its target fixed point. Ladders are warm started: the converged block at one evolution time is re-indexed onto the longer temporal lattice of the next and extended with a small random tail, with the bond cap ramped in steps of two up to χ and the truncation cutoff lowered from 10^{-8} to 10^{-10} after forty iterations; initialising consecutive times from the same subspace also improves continuity in the eigenvalue selection. When a $k = 4$ block is occupied entirely by one near-degenerate cluster, leaving the subleading eigenvalue required by Eq. (7) outside the computed subspace, the block size is escalated as $k = 4 \to 6 \to 8$, initialising each larger calculation from the converged smaller one. Either truncation scheme of Section 3.4 can be used, with the RTM as the default and the RDM as a fallback and control. Every converged (p, T) point is cached immediately and the latest converged block checkpointed, so an interrupted calculation loses at most the current point and restarts warm after resubmission.[this is okayish, but reades very dense in just a single paragraph, it would be very nice to explain it with more space in page since we do not have anymore restriction of space and pages in the appendix; also apply the comments of other sections: simple explanations, no assumptions on the reader knowing anything etc...]

F.6 Selecting the dominant eigenvalue

[this explanations seems not to add anything important to what we have already explained in the main text, so i doubt we should have an explicit subsection here explaining the same things without adding anything new... maybe there are some little details here we have not mentioned in the main text, but not enough for it to have a whole subsection, so maybe we can add small clauses in the main text or in other subsections, what do you think? propose changes and suggestiongs; also this could be done because i am thinking of reworking the whole structure of this section, which right now is not very clear or efficient given the rest of the thesis text] As stated in Section 6.2, we select the eigenvalue of largest modulus and use continuity with the preceding time point to resolve differences below one per cent. If the selected $|\mu_0|$ changes by more than a prescribed fraction relative to the previous accepted value, the point is flagged as potentially unconverged. The stored spectrum is then reanalysed by restricting the candidates to a five-per-cent window around the previous modulus and selecting the dominant eigenvalue within that interval. The reference is updated sequentially from the last accepted point. For the case with an independently converged comparison, this correction agrees with the repeated calculation to 3×10^{-5} relative. Subleading eigenvalues are ordered by their phase distance from μ_0 and matched continuously to the conformal tower.

When a numerical limitation appears, we repeat representative calculations at half the bond dimension, with a tighter truncation cutoff, and with RDM instead of RTM truncation. We interpret a feature as intrinsic to the transfer matrix only when it persists under these changes.[have we done this at all? it does not ring a bell to me, so we should not state things we have never done in practice]

F.7 Why eigenvalues outlive eigenvectors

The different numerical stability of eigenvalues and eigenvectors follows from first-order perturbation theory for non-Hermitian matrices [54]. We perturb $\mathcal{E}$ by $\delta\mathcal{E}$ and expand the perturbed right eigenvector as $|R_i\rangle + \sum_{j\neq i} a_{ij} |R_j\rangle$. Projection onto $\langle L_i|$ gives the eigenvalue shift,

$$\delta\mu_i = \frac{\langle L_i|\, \delta\mathcal{E}\, |R_i\rangle}{\langle L_i|R_i\rangle}, \tag{F.4}$$

while projection onto $\langle L_j|$ for $j \neq i$ gives the eigenvector-mixing coefficients,

$$a_{ij} = \frac{\langle L_j|\, \delta\mathcal{E}\, |R_i\rangle}{(\mu_i - \mu_j)\, \langle L_j|R_j\rangle}. \tag{F.5}$$

Equation (F.4) contains no spectral gap, so the eigenvalue remains stable while the overlap $\langle L_i|R_i\rangle$ remains finite. By contrast, Eq. (F.5) contains the denominator $\mu_i - \mu_j$. As two eigenvalues approach, their eigenvectors become increasingly sensitive to perturbations and the de-mixing of individual block members becomes ill conditioned.[this is the subsection i want you to stress test about the more: the derivation of D4 and F5 is not whatsoever clear and it assumes many things, it is a lot of claims which i am not sure whether they are right or not because the derivaiton or string of thought is not clearly explained, also, there is also the thing that after the analysis the other claude code session did, we know that the rigidity is not behind why the entropy route fails (which is one of the questions the appendices aims to answer or analyze with numerical data and results, and right now it is not clearly displayed or centered about that questions (i want to give the implementation details of our algorithm which are interesting and important, but also justify why the eigenvector path of the algorithm can fail too, with numerical data justifying our claims and with honesty on what is clear or what needs more analysis or research in the future))]

The overlap appearing in Eq. (F.4) is quantified by the phase rigidity of a biorthogonal pair,

$$r_j = \frac{|\langle L_j|R_j\rangle|}{\|L_j\|\,\|R_j\|}, \tag{F.6}$$

which is the normalised overlap of the left and right eigenvectors associated with the same eigenvalue. It equals one for a Hermitian matrix and approaches zero as their biorthogonal overlap decreases. A small rigidity therefore increases the sensitivity of an eigenvalue associated with an individual eigenvector[is this really true? also, if we have found out that rigidity does not tell us much, is it still worth to keep it in the thesis?]. The leading invariant subspace can nevertheless remain well conditioned when it is separated from the rest of the spectrum. The eigenvalue-only mode exploits this distinction: it propagates an orthonormal basis of the leading subspace and obtains the Ritz values from the projected pencil without assigning each basis vector to a particular eigenvector. Its improved stability arises from avoiding the ill-conditioned de-mixing required by the eigenvector calculation.[this last phrase is good, but i do not thing it belongs here but somewhere else] [note apart: all changes we make in the appendix should also be reflected in the main text so that no claims in the main text are false or missing after a change that we apply to the appendix] [also: we should keep the appendices as simple as possible as in the main text, so that only the strictly necessary and useful parst are left in the appendix, and reduce all the noise or things that are not that important anymore to the main text and completeness and self containess of the text clear and simple and easy to follow for a reader who is not familiar with our work or whatever we explain or mention]

The ratio $|\mu_1/\mu_0|$ controls the suppression of the first subleading contribution in Eq. (9), but it does not determine convergence of the block iteration. Convergence instead depends on the separation between the complete leading subspace and the remainder of the spectrum. Near a degeneracy within the retained block, $|\mu_1/\mu_0|$ can approach one while the subspace remains well separated and its Ritz values continue to converge.[this gets repeated with something we have already explained somewhere else, right?]

The gap denominator in Eq. (F.5) does not, however, explain why the conditioning worsens with the coupling. At short evolution times the rotated transfer operator is small enough for exact diagonalisation, which gives the conditioning of the full eigenvector basis without any tensor-network truncation[can you give more details on how we do this? so that it is more clear]. Table 5 shows the result at matched matrix dimension: the condition number of the eigenvector basis grows by seven to nine orders of magnitude between the integrable point and the interacting couplings, while the leading spectral gap is comparable or larger there and the rigidity of the dominant pair remains of order one. The ill conditioning is therefore a property of the interacting transfer matrix itself rather than a consequence of an approaching degeneracy, and the mechanism that sets it remains open[this is a really interesting result, we should rework the subsection to make an emphasis on this result so that the appendix is simpler and the main take away idea is cleanly delivered (no noise from explanations or derivations of quantities that later on are not that important at the end and that are there because before they were a candidate from the superseded analysis in the wrong construction we had for months)]. The same calculation bounds the eigenvalue sensitivity from above: first-order shifts under perturbations of the size introduced by truncation lie between one and four orders of magnitude below the bound $\|\delta\mathcal{E}\|/r_0$, so the eigenvalues are more stable than Eq. (F.4) alone guarantees[not very clear this last idea...].

p	T	dimension	$\lvert\mu_0 - \mu_1\rvert$	r_0	cond(V)
0	3.5	2187	0.21	0.12	3.0×10^2
0.1	1.5	2197	0.30	0.16	2.2×10^{11}
0.3	1.5	2197	0.19	0.17	2.1×10^{10}
0.5	1.5	2197	0.14	0.19	9.6×10^9

Table 5: **Conditioning of the exact eigenvector basis.** Exact diagonalisation of the rotated transfer operator at matched dense dimension, without cooling sites and at $\delta t = 0.5$. The fourth column is the distance between the two leading eigenvalues, r_0 the rigidity of the dominant pair from Eq. (F.6), and $\mathrm{cond}(V)$ the condition number of the full eigenvector matrix. The conditioning is set by the coupling, not by the gap.

F.8 Seed ensembles and the limits of the entropy route

[this is a very important section, so we should simplify the appendix so that the main ideas we want to convey to the reader are clearly delivered and presented, and there are no unnecessary long appendices with ideas that add little or nothing to our analysis or to the main idea... this rework needs to be analysed in great detail, propose different pahts and i will review everything and aprove what i see apropiate to keep or to drop (concerning to all the sections we are reviewing in this round)]

The loss of reliability cannot be attributed to a fixed threshold in the phase rigidity. The rigidity of the dominant pair decreases geometrically with T and does so more rapidly as the NNN coupling grows, yet its value at the last usable time spans four orders of magnitude across the couplings: at $p = 0.1$, $T = 13$ every one of ten independent runs returns a physical plateau at a rigidity of 8×10^{-10}, whereas at $p = 0.5$, $T = 6$ all ten fail at a rigidity three orders of magnitude larger. Within a single coupling the rigidity falls monotonically while the failures are intermittent. Low rigidity therefore accompanies the loss of reliability but does not determine it.[cite where the numbers you are mentioning are coming from]

To characterise the failures we repeated the single-vector calculation[using the defaults itransverse power method...] from independent random seeds at every accessible evolution time: twenty seeds per time at $p = 0$ and ten elsewhere, with the seeds recorded so that every run is reproducible[the recording of the seeds is maybe a comment we can avoid mentioning because it is superflous for the discussio in the text?]. A run counts as failed when its imaginary plateau cannot be a Rényi-2 plateau at all; since any such band is a choice, Table 6 reports each rate under a loose and a tight band around $\pi c/16$. The two agree except at the margins, and never disagree about whether a time is usable. Failure is a property of the individual run, not of the evolution time: the failed runs return plateaus that are unphysical, while the surviving runs at the same time agree with the neighbouring times. The failure probability rises with T at every coupling and its onset moves to earlier times as the coupling grows, from $T = 18$ at the integrable point to $T = 4$ at $p = 0.5$[is this really true?]. At the last accessible times the two effects combine: at $p = 0.3$, $T \geq 9$ the surviving runs themselves spread over a factor of two and a half, so beyond the failure onset an ensemble is required not only to identify the failures but to bound the surviving values[not very clear here...].

The stopping condition of the iteration carries no information about these failures. Under the RTM truncation the change in the leading eigenvalues settles above the conver-

p	T	failed (loose)	failed (tight)	plateau of surviving runs
0	8–17	0/20	0/20	0.1252–0.1137
0	18	1/20	1/20	0.1144
0	20	0/20	1/20	0.1146
0	22	2/20	2/20	0.1080
0	24	4/20	6/20	0.1205
0.1	11	2/10	2/10	0.1199
0.1	12	0/10	2/10	0.1172
0.1	13	0/10	0/10	0.1158
0.3	4–6	0/10	0/10	0.1333–0.1264
0.3	7	2/10	2/10	0.1270
0.3	8	4/10	5/10	0.1184
0.3	9	6/10	7/10	0.1054
0.3	10	6/10	7/10	0.1007
0.5	3	0/10	0/10	0.1354
0.5	4	3/10	3/10	0.1351
0.5	5	7/10	7/10	0.1310
0.5	6	10/10	10/10	–

Table 6: **Failure rate of the single-vector iteration against evolution time.** Independent random seeds per evolution time; a run fails when its imaginary plateau falls outside a band around the asymptotic value $\pi c/16 = 0.098$, with the loose band $(0.05, 0.20)$ and the tight band $(0.07, 0.15)$ showing the sensitivity to that choice. Consecutive times with no failures are grouped. The last column is the median plateau of the surviving runs.

gence tolerance and stops falling at essentially every evolution time beyond $T \approx 10$, for correct and failed runs alike, because the truncation continually reinjects noise of a size set by its cutoff. Table 7 repeats a subset of times with the RDM scheme, which truncates the left and right states on their own density matrices: the iteration then satisfies the same tolerance at every time tried, up to $T = 24$ at the integrable point, and reproduces the RTM plateau. That floor therefore belongs to the RTM truncation rather than to the transfer matrix, and the agreement between the two schemes verifies the truncation where both were run. The RDM scheme costs an order of magnitude more at every time compared, which is why the RTM remains the production choice.[this is one of the key analysis: rdm vs rtm, good]

Three questions remain open. The mechanism by which the coupling controls the conditioning of the eigenvector basis is not identified. No single-run criterion separates a degraded surviving run from a healthy one near the failure onset; only repetition reveals the distinction. And the failure statistics of the block iteration used for the spectral route have not been measured; the spectral windows of Table 1 rest on the phase-increment criterion of Section 6.2.

[vey good, but i feel that the main idea is not very clear... we are discussing in this app sec the implementation details, and all the kwargs the algorithm we used for the simulation uses; so it is important to specify in a clear way all the kwargs we have available, and also what the kwargs imply: what happens when we change the max bond dim? the cutoff? re seeding? warm vs cold starting? cutoff? what happens if i change the beta0 cooling sites? rdm vs rtm? trotter error with different dt? the mode: eigen values vs the eigenvector route?... all of these are really important details we have to take into account and justify: the main idea in the main text, and the technical details and numerical validations in the appendix. right now all of these questions and specifications are mostly or partly

p	T	plateau		stopping		time (s)	
		RTM	RDM	RTM	RDM	RTM	RDM
0	8	0.1252	0.1248	converged	converged	12	147
0	12	0.1191	0.1211	no improvement	converged	53	785
0	14	0.1186	0.1185	no improvement	converged	98	1466
0	17	0.1137	0.1162	no improvement	converged	177	2627
0	20	0.1146	0.1145	no improvement	converged	283	5177
0	22	0.1080	0.1134	no improvement	converged	402	5125
0	24	0.1205	0.1126	no improvement	converged	460	7300
0.1	11	0.1199	0.1207	no improvement	converged	1316	43692

Table 7: **The two truncation schemes at the same evolution times.** RTM entries are medians over the seed ensembles of Table 6 (surviving runs only); RDM entries are single runs, except $T = 17$ where three seeds give 0.1153–0.1165. Stopping records whether the iteration met the 10^{-6} tolerance or was stopped after 150 iterations without improvement. Times are medians per run at identical settings.

explained already in the appendix, but not displayed in a clear way or organized: it is pretty scrambled and the main idea to take home for a reader not familiar with our project or interested to know more is not cleat whatsoever; this is the whole point of the rework we are carrying out]

G Fitting conventions and supporting profiles

[i do not think the title fits the content of the section: we present complementary analysis and plots that do not fit in the main text like the comparison between fixed and free boundary conditions, the entropy plots that did not enter the main text for lack of space, and the fit windows... i also think it would be useful to add here a similar analysis as it is done in bou commas 2026 in which they test whether they recover the expected boundary dimensions and tower states according to the type of fixed-fixed, free-free, fixed-free etc... right now in the main text we reference to the boucomas2026 paper but if we replicate the results here, we could just mention the appendix which would make our claims more self contained. i want to keep consistency with how we built the appendix of the mpo construction E, which i have rewritten in my own voice, and apply the same formula here]

[list of changes: 1. we can expand G1 with the extra analysis of different boundary choices as discussed in the previous comment 2. both in the main text and appendices, the central charge readings from the temporal entropies are left as open because the cluster results are still running in the cluster; however, we have part of the data already, so analyse it and tell me whether we could use it for giving a provisional number in the table and also in the text whenever possible (save in permanent memory that these readings are provisional until more data comes from the cluster). with this said, an additional analysis we could also add the analysis of the reading of the central charge from the real part of the temp entropy: right now in the main text we say that we do not simply do it becuse the reading is more complex and elaborated than directly reading from the im plateau of the temp entropy, and this is a bit vague; what i propose is to leave in the main text as the main reading is that of the im plateau, and refer to the appendix for the calculations of the central charge from the real part which is a bit more elaborated: this way we do not sound vague and save space in the main text, while doing all the analysis and discussion in the appendix where we have no restriction in the length of the text. this new analysis has to be a mix of the appendix section in which carignano and tagliacozzo analyse the loschmidt echo and discuss about it, and also the corrections done in the boucomas paper, enter plan mode and make a throughout analysis of how to properly etract the central charge from the real part of the temp entropy and apply the technique (which you should explain in detail in a simple and clear way how we do every step) to our data to see whether we can extract c=0.5 and if not, why 3. i am no very convinced about how we determinethe fitting window, right now it is not very clear; i suggest to deliver a more simple, clear and easy to follow explanation/discussion about it, and also to update the argument to the current state of the art of the thesis given that i think this was written before the new analysis we have done in the last week; 4. i like how the branch constant is explained; make sure it is accurate to what we actually do numerically as a check 5. in other claude code session(read CONTROLS.md for summary), we have made a throughout analysis of all the numerical controls of most parameters that we use in the simualation... read it and also app F to make sure that the paragraph we talk about numerical controls does not repeat other ideas we have previously discussed, and make sure all claims are referenced correctly and displayed in a rigorous way with numerical evidence (instead of how it is now displayed in a vague and confusing way). for example, last night in that session we made an analysis of the trotter effect on our simulations which could supersed the analysis made here (or maybe it already explains it, although if that is the case i do not know if this is the best place to discuss it? analyse it and propose options) 6. we mention many numerical controls here, but i do not know if they belong to app F or G, what is the best option? i think app F contains other numerical controls too, but i am not very sure 7. last paragraph is very vague: it presents the akaile information criterion, but does not explain what it is, where does it come from (refs?) and how we use it; also the explanation is not very clear, it just displays results but not in a very clear way]

This appendix collects the conventions behind the numbers quoted in Section 7, together with the measurements that support them but are not shown there: the boundary exponents at the integrable point, the entropy profiles at the remaining couplings, and the fitting windows and controls.

G.1 The boundary exponents at the integrable point

The leading boundary exponent distinguishes the two boundary conditions of Section 2.2. At the integrable point, fitting Eq. (7) to the first spectral gap over $T = 3$–9 gives $x_1 = 0.499$ for the free boundary state $|X+\rangle$ and $x_1 = 1.988$ for the fixed state $|Up\rangle$, in agreement with the exact values $\frac{1}{2}$ and 2. Figure 19 shows both.

G.2 The entropy profiles at stronger coupling

Section 7.1 shows the entropy profiles at $p = 0.1$. Figure 20 gives the same comparison at the remaining couplings, each to the longest evolution time reached. The real part follows the chord curve at $c = \frac{1}{2}$ throughout, and the imaginary part is constant across the interior of the strip and lies above $\pi c/16$ by the finite-time correction, exactly as at the two couplings shown in the main text. The number of evolution times available falls as the coupling grows, since the cost of each one rises with p; at $p = 0.5$ only four are available, which is too few to constrain the extrapolation.

G.3 Fit windows and the controls behind the quoted numbers

The central charges in Table 1 are obtained from least-squares fits. We specify here the fitting windows, the fixed parameters, and the numerical controls used to assess their

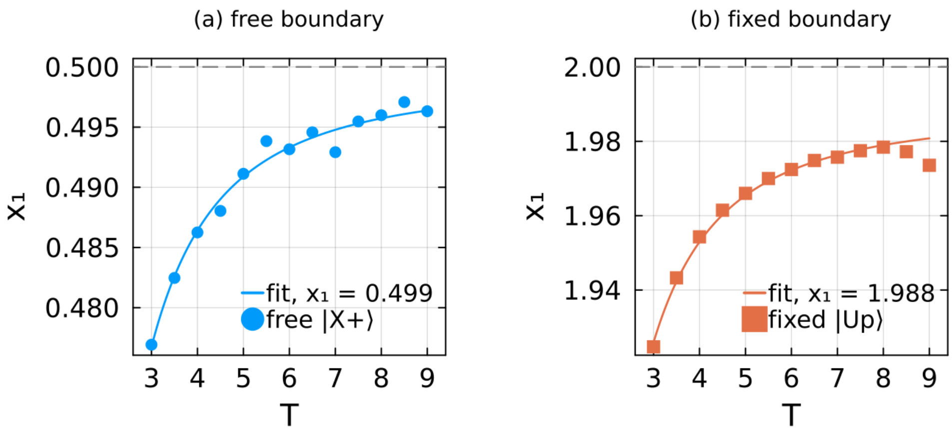


Figure 19: **The boundary exponent at the integrable point.** The first spectral gap converted into a dimension, $x_1 = vT\,\mathrm{Im}(\lambda_1 - \lambda_0)/\pi$, against T for the free and fixed boundary states. Panel (a) is the free boundary state $|X+\rangle$ and panel (b) the fixed state $|Up\rangle$, side by side, each on its own scale. Dashed lines mark the exact values $\frac{1}{2}$ and 2. Both approach their target from below as T grows, and the fits of Eq. (7) over the same thirteen evolution times give 0.499 and 1.988. The window here is shorter than the one behind Table 1, since it is cut to the range the fixed ladder covers so that the two boundary conditions are fitted over identical times; this is why the free value differs slightly from the one quoted there.

stability.

The start of the fitting window is not unique. Moving it to later times reduces the number of data points and increases the fitted central charge because the $1/T$ and $1/T^2$ contributions in Eq. (6) become difficult to distinguish over a short interval. Varying the starting time changes c by 0.014 at $p = 0$, 0.024 at $p = 0.1$, and 0.21 at $p = 0.3$. We report the widest window that satisfies the consistency criterion, since it contains the most data and best separates the correction terms. The upper end is set by the loss of a constant phase increment in μ_0. At $p = 0.5$, for example, the increment holds to a few parts in 10^4 across the fitted window and becomes irregular beyond it. This criterion determines the windows in Table 1.

We fix the branch constant in Eq. (6) rather than fitting it. Its value follows from $\lambda_0 = \log(-\mu_0)$, and unconstrained fits over sufficiently long windows give $B/\pi = -1.000$ at every coupling. Fixing it removes a parameter that short windows cannot determine. The phase is unwrapped over the complete time series before selecting a fitting window; otherwise the absolute branch information would be lost.

We use three numerical controls. Increasing the bond dimension to $\chi = 128$ changes the imaginary plateau by approximately 2% and $|\mu_0|$ by less than 10^{-4}, but shifts the real-part chord slope by up to 0.12. This motivates extracting the entropy result from the plateau. Halving the Trotter step while doubling N_β to keep β_0 fixed changes the plateau by less than 1% within the reliable window. Near the window boundary the comparison between steps is dominated by the seed dependence of Appendix F.8, so no Trotter effect can be attributed there. Repeating one time point with five independent random seeds produces a 3% spread in the plateau and a 12% spread in the chord slope, whereas $|\mu_0|$ agrees within 1.3×10^{-4}. The spectral estimates are therefore substantially less sensitive to the random seed. Finally, the sound velocity enters linearly, and its extrapolation gives 1.99994 at the integrable point, compared with the exact value 2.

A fourth control concerns the truncation cutoff on the block path. The eigenvalues are insensitive to it: across cutoffs of 10^{-8}, 10^{-10} and 10^{-12}, and across independent seeds, the

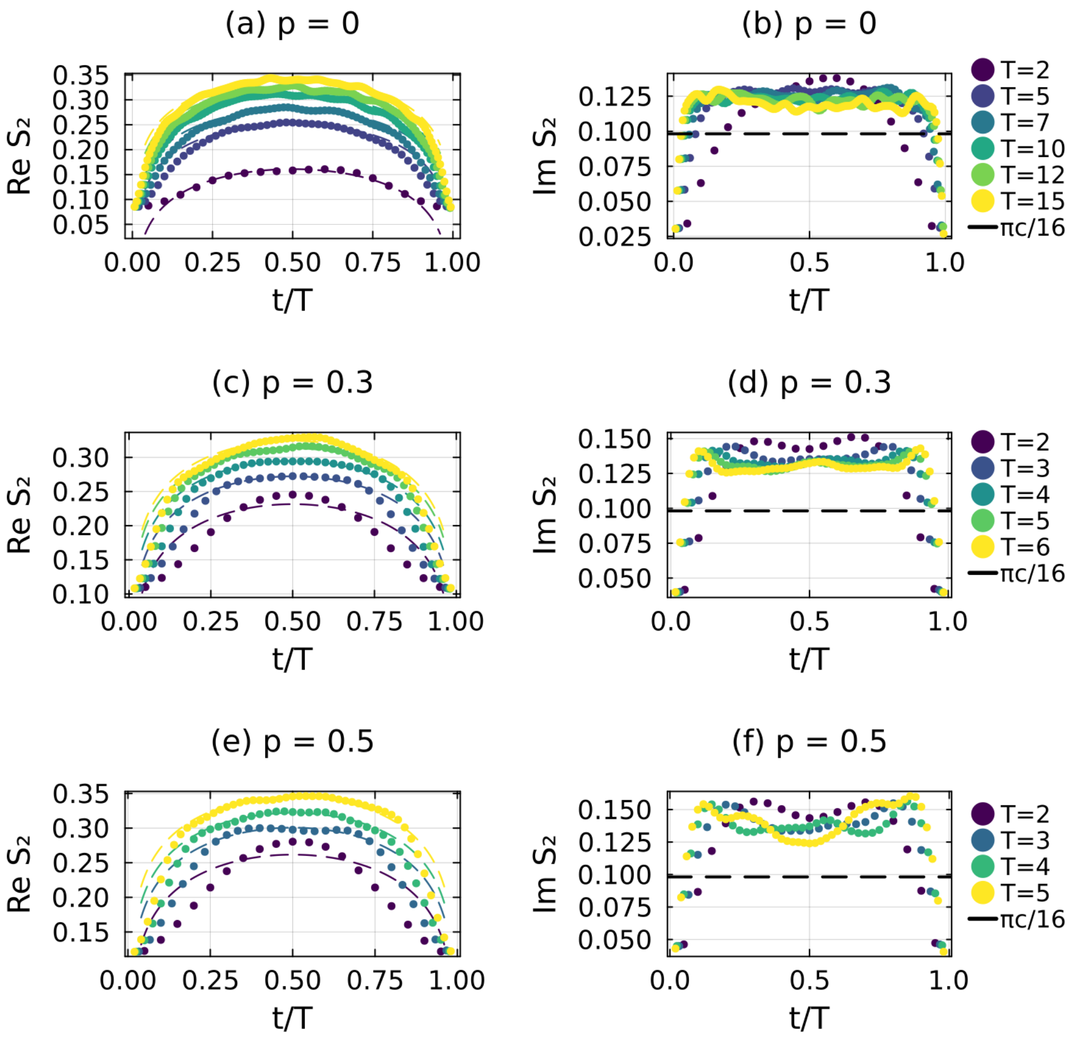


Figure 20: **The generalized temporal entropy at the remaining couplings.** The comparison of Figure 8 at $p = 0$, 0.3 and 0.5. Each row shows up to six evolution times, evenly spaced across the range the calculation reached. (a, c, e) $\mathrm{Re}\, S_2$ against t/T, with the dashed curves showing Eq. (5) at $c = \frac{1}{2}$ and the non-universal constant matched to each profile. (b, d, f) $\mathrm{Im}\, S_2$, with the dashed line at $\pi c/16 = 0.098$ for $c = \frac{1}{2}$. The separation between the data and that line is the finite-time correction, as in Figure 8.

leading modulus gap is unchanged to five decimal places and the plateau moves by less than 1.5%. What the cutoff changes is how the iteration terminates. At its production value of 10^{-8} the change in the leading Ritz values usually settles above the convergence tolerance, whereas tighter cutoffs typically meet it, but the effect depends on the random seed in both directions, and at the integrable point with fixed boundary conditions every cutoff converges within a few tens of iterations. The cutoff therefore influences the convergence path rather than the answer, consistent with the role of the truncation noise floor described in Appendix F.8.

The entropy-route central charges of Section 7.1 also depend on the finite-time correction model. We compare a constant, $a + bT^{-1/2}$, $a + b/T$, and $a + b/T + c/T^2$ using the corrected Akaike information criterion. Over $T = 2$–24, the constant model is disfavoured by a difference of 21. The $T^{-1/2}$ and two-term models differ by less than 1 and give $c = 0.506$ and 0.511, respectively. Over the shorter interval $T \leq 11$, the data do not distinguish the four models, demonstrating the need for the extended time series.